



# General Engineering

## Final Project Report Cover Sheet

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# University of Sheffield

General Engineering Final Year Project

## The application of regularisation to common datasets in machine learning

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# Abstract

Regularisation is a process used to treat linear inverse problems of the form  $\mathbf{Ax} = \mathbf{b}$  by deliberately introducing bias for a significant increase in stability. However, regularisation is commonly treated as a “black-box” default that is blindly applied in machine learning — this is problematic because its application is not benign. Unnecessarily regularising a system introduces bias without the need for added stability and suppresses meaningful features, while silently masking this misapplication by returning a finite output. This project explored the mathematical criteria that necessitate regularisation and applied these conditions to common machine learning datasets.

The condition that determines whether regularisation is required is the qualitative shape of the L-curve. Testing was conducted on the ill-posed Tecator dataset and the well-posed California housing dataset to determine if regularisation was required, and if so, how to select the optimal model. The importance of a diagnostic step prior to application was emphasised, as this helps inform model decisions. The lasso technique yielded the most accurate and stable predictive model whilst using the fewest features, a result that was validated with known physical data. Furthermore, the L-curve was established to be a more robust parameter selection method than the GCV. Ultimately, this project showed the dangers of default application of regularisation and outlined a proposed automated diagnostic step in an attempt to solve this problem for future use in machine learning pipelines.

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# Nomenclature

## Roman Letters

| Symbol                      | Description                                         |
|-----------------------------|-----------------------------------------------------|
| $\mathbf{A}$                | System matrix                                       |
| $\mathbf{A}^\dagger$        | Moore-Penrose pseudoinverse of matrix $\mathbf{A}$  |
| $\mathbf{A}_k$              | Truncated system matrix                             |
| $\mathbf{b}$                | Observed data vector                                |
| $\bar{\mathbf{b}}$          | Unperturbed observed data vector                    |
| $b_i$                       | Actual value for the $i^{\text{th}}$ observation    |
| $\hat{b}_i$                 | Predicted value for the $i^{\text{th}}$ observation |
| $c_i$                       | Fourier coefficients                                |
| $\mathbf{e}$                | Gaussian noise vector                               |
| $f_i$                       | Filter factors for regularisation                   |
| $G(\lambda)$                | Generalised Cross-Validation (GCV) function         |
| $\mathbf{I}_m$              | Identity matrix, $m \times m$                       |
| $\mathbf{I}_n$              | Identity matrix, $n \times n$                       |
| $i^*$                       | Intersection index of signal and noise components   |
| $K$                         | Number of cross-validation folds                    |
| $k$                         | Truncation index                                    |
| $k_{\text{opt}}$            | Optimum truncation index                            |
| $m$                         | Number of observations (rows in $\mathbf{A}$ )      |
| $n$                         | Number of features (columns in $\mathbf{A}$ )       |
| $p$                         | Digits of precision lost due to condition number    |
| $\mathbf{R}$                | Correlation matrix                                  |
| $r$                         | Rank of matrix $\mathbf{A}$                         |
| $\mathbf{U}$                | Orthogonal matrix of left singular vectors          |
| $\mathbf{u}_i$              | Individual left singular vector                     |
| $\mathbf{V}$                | Orthogonal matrix of right singular vectors         |
| $\mathbf{v}_i$              | Individual right singular vector                    |
| $\mathbf{x}$                | Vector of model parameters                          |
| $\mathbf{x}_{\text{enet}}$  | Elastic net regularised solution                    |
| $\mathbf{x}_{\text{lasso}}$ | Lasso regularised solution                          |
| $\mathbf{x}_{\text{ls}}$    | Least squares solution                              |
| $\mathbf{x}_{\text{reg}}$   | General regularised solution                        |
| $\mathbf{x}_{\text{tikh}}$  | Tikhonov regularised solution                       |
| $\mathbf{x}_\lambda$        | Regularised solution                                |

## Greek Letters

| Symbol                   | Description                                      |
|--------------------------|--------------------------------------------------|
| $\alpha$                 | Mixing parameter for elastic net                 |
| $\gamma_i$               | Ratio of Fourier coefficients to singular values |
| $\delta c_i$             | Noise contribution to the Fourier coefficients   |
| $\epsilon_{\text{mach}}$ | Machine epsilon                                  |
| $\eta$                   | Effective condition number                       |
| $\kappa$                 | Condition number                                 |
| $\lambda$                | Tikhonov regularisation parameter                |
| $\lambda_{\text{opt}}$   | Optimum regularisation parameter                 |
| $\mu$                    | Lasso regularisation parameter                   |
| $\sigma_i$               | Singular values of matrix $\mathbf{A}$           |
| $\sigma_{\text{noise}}$  | Standard deviation of Gaussian noise             |
| $\Sigma$                 | Diagonal matrix of singular values               |
| $\Sigma_k$               | Truncated diagonal matrix of singular values     |

## Abbreviations

| Term  | Description                                     |
|-------|-------------------------------------------------|
| BLUE  | Best Linear Unbiased Estimator                  |
| DPC   | Discrete Picard Condition                       |
| GCV   | Generalised Cross-Validation                    |
| IQR   | Interquartile Range                             |
| Lasso | Least Absolute Shrinkage and Selection Operator |
| MSE   | Mean Squared Error                              |
| NaN   | Not a Number                                    |
| NIR   | Near-Infrared                                   |
| OLS   | Ordinary Least Squares                          |
| RMSE  | Root Mean Square Error                          |
| SVD   | Singular Value Decomposition                    |
| TSVD  | Truncated Singular Value Decomposition          |
| VIF   | Variance Inflation Factor                       |

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# 1 INTRODUCTION

## 1.1 Background and Motivation

Regularisation is a process which is often applied when data are analysed to impose stability and robustness on the model. This should be applied to problems that are either highly unstable (meaning the output is extremely sensitive to input noise), lack a solution, or lack a unique solution — these problems are described as being ill-posed. It does this by solving a “neighbouring” well-conditioned problem, which inherently introduces some amount of error; therefore, regularisation can be thought of as deliberately trading a small amount of error for greatly improved stability. At the core of this project is the linear inverse problem that can be represented simply by the equation  $\mathbf{Ax} = \mathbf{b}$ . Key applications of inverse problems include image deblurring (shown in Figure 1.1), inverse kinematics of a robot arm, a control system attempting to balance two stacked spheres, and linear regression. Linear regression, where the goal is to model the relationship between a target and its features, is the real-world application that this project focuses on.



Figure 1.1: Example of an inverse problem — image deblurring.

The motivation for this project is that regularisation is often blindly applied and rarely justified. This is an issue because its use is not benign; it introduces unnecessary error without the need for added stability, or in some cases, forces a physically inaccurate model by actively suppressing valid features. Commonly in machine learning, regularisation is treated like a “black-box” prerequisite which is often applied by default using machine learning libraries in Python. This is frequently done without questioning whether this step is needed because the theory underpinning regularisation is rarely understood. This is dangerous because the regularised solution is always finite — it does not return ‘NaN’ (not a number) or infinity, which silently masks the fact that it should not have been applied. This report details the theory behind regularisation and verifies key concepts through testing before applying the theory to a real-world dataset, showing the steps that must be taken to fully understand the problem and ascertain the quality of a model and the level of confidence in its predictive results.

There are two types of linear systems that regularisation can be applied to: synthetic problems and real-world datasets. The synthetic problems provide a clean environment where the true solution is known, making it possible to directly measure accuracy, which allows objective comparison between methods. Also, these synthetic problems give complete control over the noise level, ill-posedness, and problem size, which makes a wide variety of tests possible that would otherwise be impossible on a real-world dataset. On the other hand, the ill-posedness of real-world datasets is not clean because it is rarely black or white — most often these datasets are neither severely ill-posed nor well-posed. Also, it is impossible to know the noise level or error in the measurement of the features, so it is important to determine whether one can trust the data; otherwise, it is impossible to know if the results are due to the method or the data. Testing on real-world datasets is vital, however, as a method that performs well on synthetic problems may rely on assumptions that do not hold and fail when applied to real data. Therefore, testing was conducted on both types of linear system.

## 1.2 Aims and Objectives

The aims and objectives of this project were defined at the beginning of the first semester as follows, to be achieved in chronological order.

- Define the mathematical conditions for which regularisation is necessary.
- Set up a software framework and implement these criteria in Matlab.
- Test these mathematical criteria on common machine learning datasets to categorise the need for regularisation, determining whether it is required.
- Verify that the regularised solution satisfies conditions on error and stability.

## 1.3 Report Structure

The remainder of this report is structured as follows. Section 2 provides a comprehensive overview of the mathematical foundations, primarily linear algebra, underpinning this project, as well as the mechanisms behind the different regularisation techniques and the methods of selecting the regularisation parameters. Section 3 details all materials used, as well as the steps taken for each test for reproducibility. Section 4 contains the results of the tests, which were conducted by following the methods stated in Section 3, as well as critical discussion about the significance of these results. Section 5 gathers the most important findings and provides suggestions for further work.

## 2 BACKGROUND AND LITERATURE REVIEW

### 2.1 Mathematical Foundations

Inverse problems in linear algebra, where observed data are used to determine the model parameters, are fundamental to machine learning. As established in Section 1.1, these problems can be represented by a system of linear equations of the form

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{m \times n}, \quad \mathbf{x} \in \mathbb{R}^{n \times 1}, \quad \mathbf{b} \in \mathbb{R}^{m \times 1}, \quad (2.1)$$

where  $\mathbf{A}$  represents the system,  $\mathbf{x}$  represents the model parameters and  $\mathbf{b}$  represents the observed data. In real-world applications, the observed data are always contaminated with errors arising from measurement, approximation, and rounding errors. Therefore,  $\mathbf{b}$  can be decomposed into the unperturbed observed data  $\bar{\mathbf{b}}$  and the Gaussian noise  $\mathbf{e}$ :

$$\mathbf{b} = \bar{\mathbf{b}} + \mathbf{e}. \quad (2.2)$$

Inverse problems are prone to ill-posedness, a concept first defined at the beginning of the 20<sup>th</sup> century (Hadamard 1902) as failing to meet any of the following criteria: the solution must exist, the solution must be unique, and the solution must depend continuously on the input. It is this third qualifier, the stability condition, that inverse problems most commonly fail since the processes they reverse are smoothing or diffusive, with a prime example being deblurring. This instability of these problems necessitates a mathematical diagnosis, which the singular value decomposition (SVD) addresses.

The SVD is a fundamental tool in regularisation because it reveals the quantitative measure of the ill-posedness of a problem. As its name suggests, the SVD factorises the matrix  $\mathbf{A}$  into three simpler matrices: two orthogonal matrices,  $\mathbf{U}$  and  $\mathbf{V}$ , and a diagonal matrix  $\mathbf{\Sigma}$ . The orthogonality of  $\mathbf{U}$  and  $\mathbf{V}$  implies that  $\mathbf{U}^T\mathbf{U} = \mathbf{I}_m$  and  $\mathbf{V}^T\mathbf{V} = \mathbf{I}_n$ . The decomposition is defined as follows:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \quad (2.3)$$

$$\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m] \in \mathbb{R}^{m \times m}, \quad (2.4)$$

$$\mathbf{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n) \in \mathbb{R}^{m \times n}, \quad (2.5)$$

$$\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] \in \mathbb{R}^{n \times n}. \quad (2.6)$$

The matrix  $\mathbf{A}$  is said to be ill-conditioned if the ratio  $\sigma_{\max}/\sigma_{\min}$ , known as the condition number,  $\kappa$ , is large (Turing 1948). This is because as  $\sigma_{\min} \rightarrow 0$ , the term  $1/\sigma_{\min}$  becomes arbitrarily large, which amplifies the noise in the input and renders the direct inversion unstable.

$$\kappa(\mathbf{A}) = \frac{\sigma_{\max}}{\sigma_{\min}} = \frac{\sigma_1}{\sigma_n}. \quad (2.7)$$

The condition number is a measure of the sensitivity of a system to errors or changes in the input, in this case, the observed data,  $\mathbf{b}$ . Computers operate with a double-precision floating-point format, which provides approximately 16 digits of precision — this explains an observation during testing. A rule of thumb with the condition number,  $\kappa$ , is that if  $\kappa(\mathbf{A}) = 10^p$ , then “one can expect to lose at least  $p$  digits of precision in solving the system  $\mathbf{Ax} = \mathbf{b}$ ” as per Cheney & Kincaid (1998), therefore leaving  $16 - p$  digits of precision.

The condition number is based solely on the SVD of matrix  $\mathbf{A}$ , and therefore, it does not take into account the observed data  $\mathbf{b}$ . This is a limitation, as the sensitivity of  $\mathbf{x}$  depends not only on  $\mathbf{A}$  but also on  $\mathbf{b}$  — this makes the condition number overly pessimistic. To account for the exclusion of  $\mathbf{b}$ , a new measure of sensitivity, the effective condition number, was introduced by Chan & Foulser (1988).

The effective condition number,  $\eta(\mathbf{A}, \mathbf{b})$ , differs from the condition number as it is a function of both  $\mathbf{A}$  and  $\mathbf{b}$ , which makes it a better predictor of the sensitivity of a system.

$$\eta(\mathbf{A}, \mathbf{b}) = \frac{\|\mathbf{A}^\dagger\|_2 \|\mathbf{b}\|_2}{\|\mathbf{x}_{\text{ls}}\|_2} = \frac{\|\mathbf{A}^\dagger\|_2 \|\mathbf{b}\|_2}{\|\mathbf{A}^\dagger \mathbf{b}\|_2} = \frac{\|\mathbf{b}\|_2}{\sigma_{\min} \|\mathbf{x}_{\text{ls}}\|_2}. \quad (2.8)$$

In (2.8),  $\mathbf{x}_{\text{ls}}$  refers to the least squares solution which is the best fit solution to a linear system of the form  $\mathbf{Ax} = \mathbf{b}$  when an exact solution does not exist or the system is unstable. This solution is found using the Moore-Penrose pseudoinverse,  $\mathbf{A}^\dagger$ , the definition of which depends on if matrix  $\mathbf{A}$  is overdetermined, square or underdetermined, as shown in (2.9):

$$\begin{aligned}
\mathbf{x}_{\text{ls}} &= \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 \\
&= \mathbf{A}^\dagger \mathbf{b} \\
&= \begin{cases} (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}, & m > n, \\ \mathbf{A}^{-1} \mathbf{b}, & m = n, \\ \mathbf{A}^T (\mathbf{A} \mathbf{A}^T)^{-1} \mathbf{b}, & m < n. \end{cases} \tag{2.9}
\end{aligned}$$

While ordinary least squares (OLS), as seen in (2.9), is the best linear unbiased estimator (BLUE) under ideal conditions as stated by the Gauss-Markov theorem (Montgomery et al. 2021), many linear systems are ill-conditioned, which is problematic. This is because OLS tries to fit the data perfectly with zero bias so the variance increases in the presence of noise. Here lies the bias-variance trade-off underpinning regularisation, for which the mean squared error (MSE) of a model can be defined as:

$$\text{MSE} = \text{bias}^2 + \text{variance}. \tag{2.10}$$

Regularisation uses filter factors  $f_i$  which introduce a controlled and small amount of bias in exchange for a severe reduction in the variance, which therefore reduces the MSE. Regularisation achieves this by making the regularised solution,  $\mathbf{x}_{\text{reg}}$ , the weighted summation of the right singular vectors,  $\mathbf{v}_i$ . These are weighted by the filter factors  $f_i \in [0, 1]$  which dampen the term  $1/\sigma_i$  which explodes as  $i \rightarrow n$  for an ill-conditioned problem. Regularisation can be explained simply in a single equation as seen in (2.11):

$$\mathbf{x}_{\text{reg}} = \sum_{i=1}^n f_i \frac{\mathbf{u}_i^T \mathbf{b}}{\sigma_i} \mathbf{v}_i. \tag{2.11}$$

The difference between the different regularisation methods is the definition of the filter factors  $f_i$ , which ideally damp the unwanted components of the SVD whilst the desired components remain unchanged.

## 2.2 Regularisation Techniques

The main regularisation techniques of interest for this report include truncated SVD (TSVD), Tikhonov regularisation, lasso (Least Absolute Shrinkage and Selection Operator), and elastic net. This selection provides a mix of different analytical and statistical regularisation techniques.

To understand the TSVD, it is important to understand the rank of a matrix, which

is the number of linearly independent rows (this is equal to the number of linearly independent columns) a matrix contains. The TSVD, which was formalised by Hansen (1987), simply reduces the rank of matrix  $\mathbf{A}$  from  $r = \text{rank}(\mathbf{A})$  to the truncation index  $k$ , and ignores the components of  $\mathbf{b}$  corresponding to the ignored components of  $\mathbf{A}$  — hence the name truncated SVD.

$$\mathbf{A}_k \equiv \mathbf{U}\mathbf{\Sigma}_k\mathbf{V}^T, \quad \mathbf{\Sigma}_k = \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0). \quad (2.12)$$

The TSVD acts as a low-pass filter for which the filter factors are simply a step function, giving the TSVD its discrete nature:

$$f_i = \begin{cases} 1, & i \leq k, \\ 0, & i > k. \end{cases} \quad (2.13)$$

Singular values with indices  $i > k$  are discarded, as these values are very small and cause the solution to explode. This means that the TSVD is computationally efficient and does not bias the singular values up to  $k$ . However, the TSVD cannot be fine tuned, since the chosen  $k$  must be an integer.

If TSVD can be considered discrete, the most common regularisation method, Tikhonov regularisation or ridge regression (Tikhonov 1963), can be considered continuous. This is because it uses the  $L_2$  norm to penalise the magnitude of the coefficients continuously:

$$\mathbf{x}_{\text{tikh}} = \underset{\mathbf{x}}{\text{argmin}} \left( \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 + \lambda^2 \|\mathbf{x}\|_2^2 \right). \quad (2.14)$$

The significance of the Tikhonov regularisation parameter  $\lambda$  is that it controls the trade-off between the stability and accuracy of the solution. It essentially acts as a spectral filter, suppressing the contributions from small singular values where noise amplification is most severe, while preserving the information contained in the dominant singular values. Tikhonov regularisation achieves this by defining the filter factors as follows

$$f_i = \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2}, \quad (2.15)$$

where  $\sigma_i \gg \lambda$  implies  $f_i \rightarrow 1$  and  $\sigma_i \ll \lambda$  results in  $f_i \rightarrow 0$ . Like the TSVD, Tikhonov regularisation is analytical — it has a closed-form solution, which was derived by setting the gradient of the cost function that is to be minimised (2.14) to zero to produce:

$$\mathbf{x}_{\text{tikh}} = \left( \mathbf{A}^T \mathbf{A} + \lambda^2 \mathbf{I}_n \right)^{-1} \mathbf{A}^T \mathbf{b}, \quad \text{for } m \geq n. \quad (2.16)$$

Tikhonov regularisation never fully eliminates any features, it only shrinks them close to zero. Due to its analytical nature, it is the most commonly used form of regularisation, though it is worth noting that this technique may still employ an iterative method for the sake of computational efficiency when dealing with large systems. Unlike the TSVD, Tikhonov regularisation allows for fine-tuning since  $\lambda$  is a continuous real number; however, this inherently biases all singular values, including the dominant low index singular values.

It is clear, therefore, that the choice of both  $k$  and  $\lambda$  is critical for which there are different approaches, covered in Section 2.3.

The lasso, or  $L_1$  regularisation, was first introduced by Tibshirani (1996), and this differs from ridge as it is non-linear (more specifically piecewise linear). This can shrink coefficients to exactly zero; thus, the lasso effectively performs feature selection and regularisation simultaneously. As its name suggests, this is achieved by using the  $L_1$  norm shown below. Note that  $\mu$  is the lasso regularisation parameter:

$$\mathbf{x}_{\text{lasso}} = \underset{\mathbf{x}}{\operatorname{argmin}} \left( \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 + \mu \|\mathbf{x}\|_1 \right). \quad (2.17)$$

Unlike ridge, the lasso has no closed-form solution, meaning it is a numerical problem since it requires an iterative process to solve, making it more computationally expensive. Like ridge, this is an example of a convex optimisation problem, which guarantees that the local minimum is always the global minimum. However, in underdetermined systems ( $n > m$ ) or in the presence of multicollinearity, which can be diagnosed using the Variance Inflation Factor (VIF), lasso “tends to select only one variable from the group and does not care which one is selected” (Zou & Hastie 2005) — this is problematic.

To address this problem, the elastic net, which is a regularisation technique that uses both ridge and lasso, was introduced by Zou & Hastie (2005). It does this in proportions controlled by a mixing parameter,  $\alpha$ , with pure ridge being  $\alpha = 0$  and pure lasso being  $\alpha = 1$ . Elastic net shares a similar sparsity of representation with the lasso; however it is particularly useful when matrix  $\mathbf{A}$  is underdetermined ( $n > m$ ), unlike the lasso.

$$\mathbf{x}_{\text{enet}} = \underset{\mathbf{x}}{\operatorname{argmin}} \left( \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 + \mu \|\mathbf{x}\|_1 + \lambda^2 \|\mathbf{x}\|_2^2 \right). \quad (2.18)$$

This describes the cost function to be minimised, which is normally specified in practice

by defining the mixing parameter,  $\alpha$  (Hastie 2009) along with the number of cross-validation folds used. The mixing parameter  $\alpha$  is a function of the two regularisation parameters:

$$\alpha = \frac{\mu}{\mu + \lambda^2}. \quad (2.19)$$

The selection of  $\mu$  and  $\alpha$  is more complex and computationally costly than selecting  $k$  and  $\lambda$  since the TSVD and Tikhonov regularisation are both analytical, which will be covered in Section 2.3.

The main relevant regularisation techniques in Section 2.2 are summarised in Table 2.1 in an easy-to-digest format.

Table 2.1: Comparison of Regularisation Techniques

| Technique   | Penalty     | Filter type | Parameter     | Primary effect              |
|-------------|-------------|-------------|---------------|-----------------------------|
| TSVD        | $L_2$       | Discrete    | $k$           | Rank reduction (discrete)   |
| Tikhonov    | $L_2$       | Continuous  | $\lambda$     | Rank reduction (continuous) |
| Lasso       | $L_1$       | Non-linear  | $\mu$         | Feature selection           |
| Elastic Net | $L_1 + L_2$ | Hybrid      | $\mu, \alpha$ | Selection & grouping        |

## 2.3 Selection of the Regularisation Parameter

The L-curve is a numerical method for determining the optimum regularisation parameters  $\lambda$  or  $k$ , which consists of a parametric plot of the solution norm  $\|\mathbf{x}\|_2$  against the residual norm  $\|\mathbf{Ax} - \mathbf{b}\|_2$  on log-log axes as the regularisation parameters  $\lambda$  or  $k$  vary. This was first used by Lawson & Hanson (1974) and further developed by Hansen & O’Leary (1993) where it was formally introduced. The L-curve serves two purposes: to identify by visual inspection (namely the presence of a distinct corner) whether regularisation is required and to determine the regularisation parameters  $\lambda$  or  $k$ . Therefore, the L-curve is a critical diagnostic tool for this project, allowing the determination of the genuine need for regularisation in machine learning datasets.

Analogous to how the L-curve provides a visual indication of ill-posedness, the Discrete Picard Condition (DPC) is a way to determine quantitatively if a stable solution is attainable. The DPC is concerned with the behaviour of the Fourier coefficients  $c_i = \mathbf{u}_i^T \mathbf{b}$  relative to the singular values  $\sigma_i$ . This concept was first introduced by Varah (1979) who observed that the Fourier coefficients must decay to zero faster than the singular values for a solution to be stable — this was later formally defined by Hansen (1990) as



the DPC. The necessity for the DPC is evident in the expansion of the unregularised solution, equivalent to (2.11) with  $f_i = 1$ :

$$\mathbf{x}_{ls} = \sum_{i=1}^n \frac{c_i}{\sigma_i} \mathbf{v}_i. \quad (2.20)$$

If the magnitude of the numerator  $|c_i|$  does not decay faster than the denominator  $\sigma_i$ , the individual terms in the sum will diverge, resulting in a solution dominated by noise. The noise floor of the singular values in the Picard plot of an ill-conditioned system as established previously in Section 2.1, is  $10^{-16}$  due to the double-precision floating-point format computers generally use.

Generalised Cross-Validation (GCV) is a statistical method for choosing the regularisation parameters in Tikhonov regularisation (2.14). Crucially, it provides an estimate without requiring prior knowledge of the noise variance (Golub et al. 1979), unlike the Discrepancy Principle which relies on an accurate estimate of the noise. This builds on the statistical concept of cross-validation first introduced by Stone (1974) as a way to measure prediction error by leaving out some data, fitting the model, and assessing the ability of the model to predict the missing data. Instead of leaving out data, GCV relies upon linear algebra to efficiently approximate the prediction error. Specifically, the chosen regularisation parameter is the one that minimises the following function:

$$G(\lambda) = \frac{\|\mathbf{Ax} - \mathbf{b}\|_2^2}{\left(m - \sum_{i=1}^n \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2}\right)^2}, \quad (2.21)$$

where  $m$  represents the number of data points (the rows of matrix  $\mathbf{A}$ ). The function  $G(\lambda)$  typically forms a convex curve when plotted against  $\lambda$ , which allows the optimum Tikhonov regularisation parameter  $\lambda_{\text{opt}}$  to be identified at the minimum. Note that GCV can also be used to find the optimum truncation index  $k_{\text{opt}}$  for the TSVD.

While the L-curve and GCV work well for the  $L_2$  norm SVD-based techniques of TSVD and Tikhonov, they are less effective for  $L_1$  norm techniques such as lasso and elastic net. Therefore, non-linear techniques typically require a more computationally expensive approach known as K-fold cross-validation (Stone 1974) since they are numerical and not analytical. K-fold cross-validation is a statistical method, like GCV, that works by randomly shuffling and then dividing the dataset into  $K$  equally sized groups, known as ‘folds’. The model is then trained on  $K - 1$  folds and then tested on the remaining fold. This process is repeated  $K$  times so that each fold is tested exactly once to reduce the likelihood of underfitting or overfitting. The optimal regularisation parameters, in this case  $\mu$  or  $\alpha$ , are the ones that minimise the MSE across all folds. It

follows that the computational cost increases linearly with  $K$ , however the total cost is also a function of the resolution and dimensions of the parameter grid. Since elastic net requires optimisation of both  $\mu$  and  $\alpha$ , this is a two-dimensional grid search which is even more demanding than the already computationally costly lasso (one-dimensional grid search).

The main relevant parameter selection methods in Section 2.3 are summarised in Table 2.2.

Table 2.2: Comparison of Parameter Selection Methods

| Method    | Category    | Computational cost | Selection criteria         |
|-----------|-------------|--------------------|----------------------------|
| L-curve   | Numerical   | Low — $O(n)$       | Corner (maximum curvature) |
| GCV       | Statistical | Low — $O(n)$       | Global minimum             |
| K-fold CV | Statistical | High               | Lowest average MSE         |

## 3 MATERIALS AND METHODS

### 3.1 Materials

#### 3.1.1 Hardware and Software Framework

All tests were run on a MacBook Air (M1, 2020) with 8 GB of memory, which comfortably met the computational requirements for this project. The software framework used on this setup to gather all results was Matlab R2022b, which was chosen for its ease of working with linear algebra and its clear and highly customisable graphs. However, the main reason Matlab was chosen was the availability of the `regtools` package by Hansen (2007), which contained functions that were constantly used for all tests, including `csvd`, `l_curve`, `gcv`, and `picard` as well as synthetic test problems. Also, the Matlab add-on “Statistics and Machine Learning Toolbox” (The MathWorks Inc. 2022) was installed for the `lasso` function.

These functions were used on the following problems in the form  $\mathbf{Ax} = \mathbf{b}$  using both synthetic and real-world data. These problems were deliberately selected to provide both an example of when regularisation should be applied, and an example of when it must not be applied as shown in Table 3.1.

Table 3.1: Matrix of datasets used for testing.

| Data category | Regularisation required | Regularisation not required |
|---------------|-------------------------|-----------------------------|
| Synthetic     | <code>shaw</code>       | <code>deriv2</code>         |
| Real-world    | Tecator                 | California housing          |

#### 3.1.2 Synthetic Test Problems

The synthetic test problems used from the `regtools` package (Hansen 2007) were `shaw` and `deriv2`, which were used to compare a severely ill-posed and mildly ill-posed problem. These are both discretisations of a Fredholm integral equation of the first kind. The `shaw` problem is a one-dimensional model of an image restoration problem by Shaw (1972), which is severely ill-posed. This problem requires that  $n$  is even; therefore,  $n = 50$  was used throughout, as this had sufficient features for visualising the DPC.

The `deriv2` problem (Hansen 2007) models the computation of the second derivative which is a mildly ill-posed problem; therefore, its singular values decay slowly to zero. For ease of comparison,  $n = 50$  was used throughout to match the size of the `shaw` problem. Note that these produce square matrices; therefore,  $n = m$ .

### 3.1.3 Real-World Datasets

The Tecator dataset (StatLib Datasets Archive 1995) was accessed via the open-source machine learning library `scikit-learn` and imported as a `.csv` file into Matlab. The Tecator dataset is the data produced from the near-infrared (NIR) spectroscopy of 240 meat samples and their associated fat, moisture, and protein contents. The 100 features are channels of NIR absorbance spectra from 850 nm to 1050 nm in 2 nm increments and are used to predict the target, which for this project was the moisture content of the meat samples. This represents a classic real-world ill-posed problem due to its high condition number and most notably high multicollinearity, which is caused by neighbouring channels overlapping and capturing broad absorption bands.

The California housing dataset, originally introduced by Pace & Barry (1997), was also accessed with `scikit-learn` and imported into Matlab as a `.csv` file. The California housing dataset has 8 features that are used to predict the house price of 20640 houses across California, which were derived from the 1990 United States Census. These features are as follows: median income in block group, median house age in block group, average rooms per household, average bedrooms per household, block group population, average number of occupants per household, latitude and longitude. This represents a classic real-world well-posed problem due to its low condition number and multicollinearity, meaning it has stable and independent features.

## 3.2 Methods

### 3.2.1 Condition Number Rule of Thumb Verification

The aim of this test was to empirically verify the rule of thumb regarding the condition number introduced in Section 2.1, which is that for a given  $\kappa(\mathbf{A}) = 10^p$ , the remaining digits of precision are  $16 - p$  (Cheney & Kincaid 1998). This was achieved using synthetic matrices with pre-defined condition numbers and calculating the relative error. For each target condition number in the range  $10^0$  to  $10^{16}$ , a square matrix  $\mathbf{A} \in \mathbb{R}^{n \times n}$  with  $n = 100$  was constructed via SVD such that  $\kappa(\mathbf{A}) = \kappa_{\text{target}}$ . This was achieved by generating singular values  $\sigma_i$  with a logarithmic distribution between 1 and  $1/\kappa_{\text{target}}$ , so that  $\kappa(\mathbf{A}) = \sigma_{\max}/\sigma_{\min} = 1/(1/\kappa_{\text{target}}) = \kappa_{\text{target}}$ . To ensure the synthetic matrices were non-trivial, the orthogonal matrices  $\mathbf{U}$  and  $\mathbf{V}$  were obtained from SVD of a random  $n \times n$  Gaussian matrix. To ensure reproducibility, the random number generator was initialised with a fixed seed using the Matlab `rng(4)` function. The true solution  $\mathbf{x}_{\text{true}}$  was defined as a vector of ones,  $\mathbf{1}_n = (1, \dots, 1)^T \in \mathbb{R}^{n \times 1}$ . The corresponding observation vector was calculated as  $\mathbf{b} = \mathbf{A}\mathbf{x}_{\text{true}}$ , which was used to obtain the least squares solution  $\mathbf{x}_{\text{ls}} = \mathbf{A}^\dagger \mathbf{b}$  using the Matlab backslash operator. The

observed and theoretical relative error was calculated with the following equations:

$$\text{Relative error}_{\text{observed}} = \frac{\|\mathbf{x}_{\text{true}} - \mathbf{x}_{\text{ls}}\|_2}{\|\mathbf{x}_{\text{true}}\|_2}, \quad (3.1)$$

$$\text{Relative error}_{\text{theoretical}} = \epsilon_{\text{mach}} \cdot \kappa(\mathbf{A}), \quad (3.2)$$

where machine epsilon is  $\epsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$ . Both the observed and theoretical relative error were plotted against the actual condition number which was calculated using the `cond` function in Matlab.

### 3.2.2 Comparison between $\kappa(\mathbf{A})$ and $\eta(\mathbf{A}, \mathbf{b})$

The aim of this test was to compare the difference between the condition number  $\kappa(\mathbf{A})$  and the effective condition number  $\eta(\mathbf{A}, \mathbf{b})$  on the `shaw(50)` problem, described in Section 3.1.2, as the composition of the observed data  $\mathbf{b}$  is altered. Once  $\mathbf{A}$  was defined using `shaw(50)`, the `csvd` function was used to decompose  $\mathbf{A}$  into  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{V}$ . For each index  $i$  from 1 to 50, only the first  $i$  left singular vectors are used to construct  $\mathbf{b}$ :

$$\mathbf{b}_i = \sum_{j=1}^i \mathbf{u}_j. \quad (3.3)$$

Therefore, as  $i$  increases, increasing small singular values are included in  $\mathbf{b}_i$ . Both  $\kappa(\mathbf{A})$  and  $\eta(\mathbf{A}, \mathbf{b})$  were calculated using (2.7) and (2.8), respectively, and then plotted against the index  $i$ . The Picard plot was also generated using the `picard` function together with  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{b}$ .

### 3.2.3 Bias-variance Trade-off Visualisation

This experiment was designed to illustrate the importance of the regularisation parameter  $\lambda$  on the stability and accuracy of the solution, demonstrating the bias-variance trade-off (2.10). The matrices  $\mathbf{A}$ ,  $\bar{\mathbf{b}}$  and  $\mathbf{x}_{\text{true}}$  were defined using the ill-posed `shaw(50)` problem described in Section 3.1.2. To be more realistic, a Gaussian noise vector  $\mathbf{e}$  with a fixed noise level of  $\sigma_{\text{noise}} = 1 \times 10^{-2}$  was added to  $\bar{\mathbf{b}}$ ,  $\mathbf{b} = \bar{\mathbf{b}} + \mathbf{e}$ , where  $\sigma_{\text{noise}}$  is the standard deviation of Gaussian noise. For reproducibility, the random number generator was set with a fixed seed using the Matlab `rng(1)` function. The matrices  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{V}$  were calculated using the compact SVD function `csvd` from the `regtools` package. The optimum regularisation parameter  $\lambda_{\text{opt}}$  was identified using  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{b}$  with the `l_curve` function from `regtools` package with the technique set to Tikhonov regularisation. The cases used to demonstrate this were high variance, optimal balance and high bias, for which  $\lambda$  was set to  $0.05\lambda_{\text{opt}}$ ,  $\lambda_{\text{opt}}$  and  $100\lambda_{\text{opt}}$ , respectively. The

regularised solution  $\mathbf{x}_\lambda$  was calculated for each case using (2.11) introduced in Section 2.1. The regularised solution for each of the cases, as well as  $\mathbf{x}_{\text{true}}$ , were plotted on a graph with  $x$  against  $i$  and the relative error for each case was calculated with the following equation:

$$\text{Relative error}_{\text{regularised}} = \frac{\|\mathbf{x}_{\text{true}} - \mathbf{x}_\lambda\|_2}{\|\mathbf{x}_{\text{true}}\|_2}. \quad (3.4)$$

### 3.2.4 Impact of Noise on Spectral Filtering and the DPC

The aim of this test was to demonstrate the impact of the level of noise on the filter factors for Tikhonov regularisation and TSVD as well as the DPC. The `shaw(50)` problem, described in Section 3.1.2, was used to define  $\mathbf{A}$  and  $\mathbf{b}$  with two different noise levels:  $\sigma_{\text{noise}} = 1 \times 10^{-2}$  and  $\sigma_{\text{noise}} = 1 \times 10^{-10}$ . Again, `rng(1)` was used to allow reproducibility. The `csvd` function from `regtools` was used to obtain the matrices  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{V}$ . The `l_curve` function from `regtools` was used to identify  $\lambda_{\text{opt}}$  and  $k_{\text{opt}}$  for each noise level by specifying the technique to be ‘Tikh’ and ‘tsvd’ respectively. The filter factors were defined for each technique using (2.15) and (2.13) respectively which were plotted against  $i$  for the two noise levels. Also, the Picard plots for each noise level were plotted using  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{b}$  with the `picard` function from `regtools`.

### 3.2.5 Parameter Selection Methods as Diagnostic Tools

The aim of this test was to see which of the parameter selection methods, L-curve and GCV, can be used to diagnose whether regularisation is required. Here, the mildly ill-posed `deriv2(50)` problem, described in Section 3.1.2, was used to define  $\mathbf{A}$  and  $\mathbf{b}$ . A fixed noise level of  $\sigma_{\text{noise}} = 1 \times 10^{-2}$  was applied to  $\mathbf{b}$  using `rng(1)`. Matrix  $\mathbf{A}$  was decomposed using `csvd` to obtain  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{V}$ . The `l_curve` and `gcv` functions were used together with  $\mathbf{U}$ ,  $\mathbf{\Sigma}$ ,  $\mathbf{b}$  and the ‘Tikh’ technique to produce the L-curve and GCV plots.

### 3.2.6 Robustness of Parameter Selection Methods

The aim of this test was to evaluate the robustness of the parameter selection methods, namely the L-curve and GCV. To do this, a Monte Carlo simulation was conducted over 1000 trials with different realisations of Gaussian noise at a fixed level of  $\sigma_{\text{noise}} = 1 \times 10^{-2}$ , using a fixed seed random number generator `rng(1)`. The matrices  $\mathbf{A}$  and  $\bar{\mathbf{b}}$  were defined using the `shaw(50)` problem. Matrix  $\mathbf{A}$  was decomposed into  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{V}$ . For each trial,  $\mathbf{b}$  was constructed using a different realisation of Gaussian noise. Matrices  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{b}$  were used with `l_curve` and `gcv` to identify the two values of  $\lambda_{\text{opt}}$ . The mean and standard deviation of  $\lambda_{\text{opt}}$  over the 1000 trials were calculated. The Matlab function `boxplot` was used to plot the boxplot for which the central line is

the median, the top and bottom edges of the box indicate the 75<sup>th</sup> and 25<sup>th</sup> percentiles, respectively, defining the interquartile range (IQR). The whiskers extend to the most extreme data points that are not considered outliers, while the outliers are indicated with the ‘x’ symbol.

### 3.2.7 Tecator Dataset Diagnosis

The aim for this diagnostic step was to gain an understanding of the dataset and primarily determine whether or not regularisation was required. Matrix  $\mathbf{A}$  was defined as the 100 features with the full 240 samples, therefore  $\mathbf{A} \in \mathbb{R}^{240 \times 100}$ . Matrix  $\mathbf{b}$  was defined as the column corresponding to the moisture content, therefore  $\mathbf{b} \in \mathbb{R}^{240 \times 1}$  — note that for the real datasets, no noise was added since there is an unknown but non-zero amount of noise in them to begin with. The `csvd` function was used to determine  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{V}$ . Both  $\kappa(\mathbf{A})$  and  $\eta(\mathbf{A}, \mathbf{b})$  were calculated using (2.7) and (2.8) respectively. The VIF, which is a measure of the multicollinearity within the dataset, was calculated using the Matlab function `corrcoef` which computes the correlation matrix  $\mathbf{R}$  and then taking the inverse,  $\mathbf{R}^{-1}$ . The maximum, mean and minimum VIF values were then calculated. The matrices  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{b}$  were used with the `picard`, `l_curve` and `gcv` functions for the ‘Tikh’ technique to generate the Picard, L-curve, and GCV plots respectively.

### 3.2.8 Comparison of Regularisation Techniques on the Tecator Dataset

The aim of this test was to compare the performance of the regularisation techniques on both the full Tecator dataset and underdetermined Tecator dataset. The underdetermined case was tested to see if the elastic net outperformed the lasso in this case as predicted by Zou & Hastie (2005). The procedure for the full and underdetermined dataset was as follows. For the full dataset, matrices  $\mathbf{A}$  and  $\mathbf{b}$  were split using an 80/20 training/testing data split into  $\mathbf{A}_{\text{train}}$ ,  $\mathbf{A}_{\text{test}}$ ,  $\mathbf{b}_{\text{train}}$  and  $\mathbf{b}_{\text{test}}$ ; therefore, there were 192 training samples and 48 testing samples. For the underdetermined dataset,  $\mathbf{A}_{\text{train}} \in \mathbb{R}^{99 \times 100}$  and therefore there were 141 testing samples. These tests had 50 trials, for which each trial used a different randomly shuffled  $\mathbf{A}_{\text{train}}$  using the `randperm` Matlab function — as always, `rng(1)` was used to ensure reproducibility. For each trial, the chosen regularisation technique was applied and the regularisation parameter was always selected using 5-fold cross-validation, except for the unregularised case for which  $\lambda_{\text{opt}} = 0$ . Since the elastic net requires a two-dimensional grid search, a discrete set of  $\alpha$  values was used to reduce the computational cost —  $\alpha \in \{0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.99\}$ . The `lasso` function (The MathWorks Inc. 2022) was used to perform the lasso and elastic net by specifying  $\alpha = 1$  and  $\alpha \in (0, 1)$ , respectively. The final weights for each channel, once calculated, were used

to find the root mean square error (RMSE) for the current trial using the following equation

$$\text{RMSE} = \sqrt{\frac{1}{m_{\text{test}}} \sum_{i=1}^{m_{\text{test}}} (b_i - \hat{b}_i)^2} \quad (3.5)$$

where  $m_{\text{test}}$  denotes the number of test samples,  $b_i$  represents the actual moisture content, and  $\hat{b}_i$  is the value predicted by the model. The mean and standard deviation of the RMSE across all 50 trials was calculated, as well as the mean number of active features and run time. Also, a regression plot of the predicted against the actual moisture content for the median trial was produced for each case.

### 3.2.9 Feature Selection with the Lasso on the Tecator Dataset

This test focuses solely on the lasso with the aim of identifying the most important features (wavelengths) that the lasso selects and how consistently they were selected. This was conducted on the full and underdetermined Tecator dataset, defined in Section 3.2.8. Again, this test had 50 trials, for which each trial used a different randomly shuffled  $\mathbf{A}_{\text{train}}$  using the `randperm` Matlab function and again, `rng(1)` was used to ensure reproducibility. The `lasso` function with  $\alpha = 1$  was used to determine the final weights for each trial by selecting  $\lambda$  via 5-fold cross-validation. Once completed, the selection frequency for each feature was calculated, which is the percentage of trials that a feature was chosen (given a non-zero weight); this was then displayed on a bar chart.

### 3.2.10 Model Convergence with the Lasso on the Tecator Dataset

The aim of this test was to plot a learning curve, again focusing solely on the lasso, to build confidence that the data was of high quality. This used the same method described in Section 3.2.8 and varied  $m_{\text{train}}$  and then recorded the mean RMSE using the following discrete set:  $m_{\text{train}} \in \{50, 75, 99, 125, 150, 175, 192\}$ . Note that all remaining samples were used for testing,  $m_{\text{test}} = 240 - m_{\text{train}}$ , so using more training samples means fewer samples for testing. This produced a learning curve plot of mean RMSE against  $m_{\text{train}}$ .

### 3.2.11 California Housing Dataset Diagnosis

Similarly to Section 3.2.7, the aim for this diagnosis was to develop an understanding of the dataset and determine whether or not regularisation was required. The California housing dataset caps the target, the median house value, at \$500k and caps the house age at 52; therefore, samples containing these values were filtered out of the dataset



— this left 18555 of the 20640 samples. Since the features have different units, each column of  $\mathbf{A}$  was min-max normalised to the range  $[0, 1]$ . Matrix  $\mathbf{A}$  was defined as the 8 features with the full 18555 samples, therefore  $\mathbf{A} \in \mathbb{R}^{18555 \times 8}$ . Matrix  $\mathbf{b}$  was defined as the column corresponding to the median house value, therefore  $\mathbf{b} \in \mathbb{R}^{18555 \times 1}$  — again, no noise was added since there is an unknown but non-zero amount of noise in it to begin with. The  $\kappa(\mathbf{A})$ ,  $\eta(\mathbf{A}, \mathbf{b})$  and VIF values were calculated using the same method stated in Section 3.2.7. The L-curve was plotted using `l_curve` with  $\mathbf{U}$ ,  $\mathbf{\Sigma}$ ,  $\mathbf{b}$  and the ‘Tikh’ technique.

### 3.2.12 Impact of Sample Size on California Housing Dataset

Similarly to Section 3.2.10, the aim of this test was to build confidence in the data, since stable results imply the data are of high quality. The number of samples used was the discrete set  $m \in \{100, 1000, 5000, 10000\}$ . For each sample size, a Monte Carlo simulation over 1000 trials was conducted, with each trial randomly shuffling the samples selected while using `rng(1)`. Matrices  $\mathbf{A}$  and  $\mathbf{b}$  were defined as they were in Section 3.2.11 on the filtered dataset. For each trial, the residual norm  $\|\mathbf{Ax} - \mathbf{b}\|_2$  and the solution norm  $\|\mathbf{x}\|_2$  were calculated over 200  $\lambda$  values from  $\lambda_{\min} = 0.1\sigma_{\min}^2 = 0.1\sigma_8^2$  to  $\lambda_{\max} = 10\sigma_{\max}^2 = 10\sigma_1^2$ . The trial-averaged mean values of  $\log \|\mathbf{Ax} - \mathbf{b}\|_2$  against  $\log \|\mathbf{x}\|_2$  were plotted to form the L-curve for each sample size.

## 4 RESULTS AND DISCUSSION

### 4.1 Empirical Verification using Synthetic Problems

#### 4.1.1 Condition Number Rule of Thumb Verification

The test to verify the condition number rule of thumb, outlined in Section 2.1, by creating artificial matrices with target condition numbers yielded Figure 4.1 and Table 4.1.

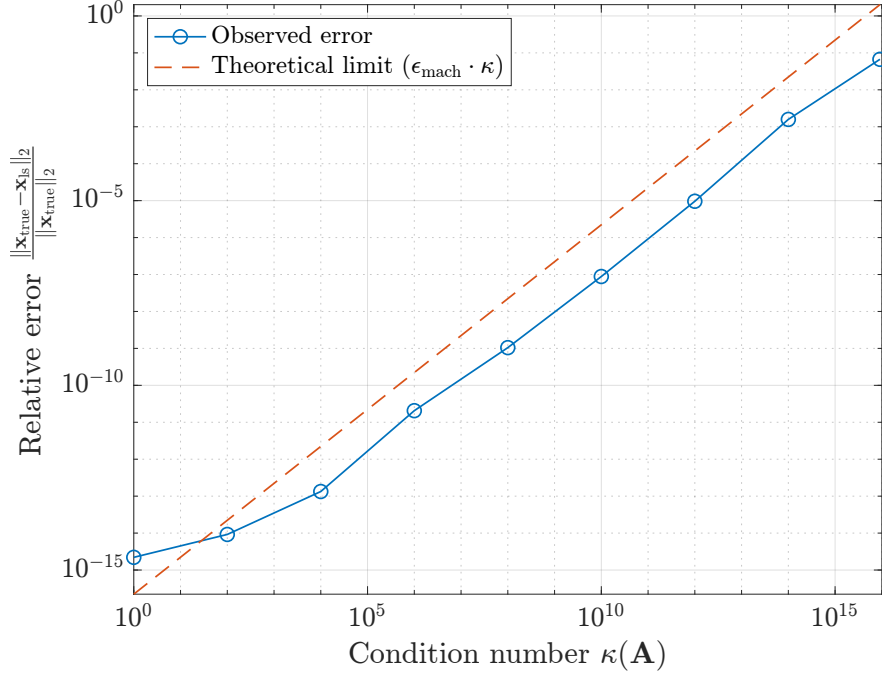


Figure 4.1: Verification of the rule of thumb relating the condition number  $\kappa(\mathbf{A})$  to the digits of precision  $p$ .

Table 4.1: Relationship between the condition number  $\kappa(\mathbf{A})$  and precision loss in the double-precision floating-point format.

| Target $\kappa(\mathbf{A})$ | Actual $\kappa(\mathbf{A})$ | Digits lost ( $p$ ) | $\frac{\ \mathbf{x}_{\text{true}} - \mathbf{x}_{\text{ls}}\ _2}{\ \mathbf{x}_{\text{true}}\ _2}$ | Digits left ( $16 - p$ ) |
|-----------------------------|-----------------------------|---------------------|--------------------------------------------------------------------------------------------------|--------------------------|
| $10^0$                      | $1.0000 \times 10^0$        | 0.00                | $2.2064 \times 10^{-15}$                                                                         | 16.00                    |
| $10^2$                      | $1.0000 \times 10^2$        | 2.00                | $9.1922 \times 10^{-15}$                                                                         | 14.00                    |
| $10^4$                      | $1.0000 \times 10^4$        | 4.00                | $1.3349 \times 10^{-13}$                                                                         | 12.00                    |
| $10^6$                      | $1.0000 \times 10^6$        | 6.00                | $2.0600 \times 10^{-11}$                                                                         | 10.00                    |
| $10^8$                      | $1.0000 \times 10^8$        | 8.00                | $1.0479 \times 10^{-9}$                                                                          | 8.00                     |
| $10^{10}$                   | $1.0000 \times 10^{10}$     | 10.00               | $8.8325 \times 10^{-8}$                                                                          | 6.00                     |
| $10^{12}$                   | $1.0000 \times 10^{12}$     | 12.00               | $9.6167 \times 10^{-6}$                                                                          | 4.00                     |
| $10^{14}$                   | $1.0004 \times 10^{14}$     | 14.00               | $1.5811 \times 10^{-3}$                                                                          | 2.00                     |
| $10^{16}$                   | $9.1022 \times 10^{15}$     | 15.96               | $6.6957 \times 10^{-2}$                                                                          | 0.04                     |

The observed error shown in Figure 4.1 consistently sits essentially parallel and an

order of magnitude below the theoretical limit, suggesting that the condition number provides an upper bound to the sensitivity of a system to noise as suspected. This trend is not followed for  $\kappa(\mathbf{A}) < 10^4$ , where the relative error actually exceeds the theoretical limit for well-conditioned matrices. This discrepancy arises because, for low condition numbers, the error is no longer dominated by the amplification of noise through  $\mathbf{A}$ . Instead, the accumulation of rounding errors in processes such as SVD dominate the relative error and mask the sensitivity to noise predicted by the condition number.

Table 4.1 shows the target and achieved condition numbers for each of the artificial matrices generated. The actual  $\kappa(\mathbf{A})$  value achieved for extremely ill-conditioned matrices, when  $\kappa(\mathbf{A}) > 10^{12}$ , deviates from the target values more with increasing  $\kappa$ . As  $\kappa$  approaches  $10^{16}$ , the smallest singular values become indistinguishable from numerical noise. At this threshold, the limits of the double-precision floating-point format prevent the precise generation of the target  $\kappa(\mathbf{A})$ , resulting in the observed deviation.

#### 4.1.2 Comparison between $\kappa(\mathbf{A})$ and $\eta(\mathbf{A}, \mathbf{b})$

The test to compare how  $\kappa(\mathbf{A})$  and  $\eta(\mathbf{A}, \mathbf{b})$  change with the number of singular vectors included in  $\mathbf{b}$  using the `shaw(50)` problem produced Figure 4.2.

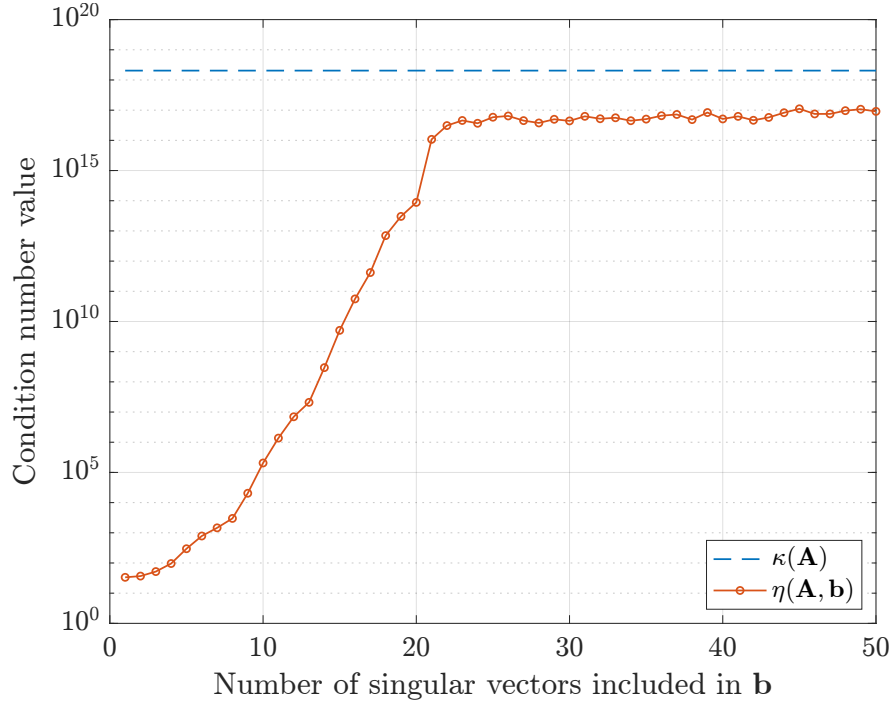


Figure 4.2: Comparison of how  $\kappa(\mathbf{A})$  and  $\eta(\mathbf{A}, \mathbf{b})$  change as more left singular vectors are included in  $\mathbf{b}$ .

As expected, the condition number did not change with  $\mathbf{b}$  since it is purely a function of  $\mathbf{A}$  and the effective condition number did change with  $\mathbf{b}$  since  $\eta$  is a function of  $\mathbf{b}$ . Additionally, the condition number value,  $\kappa(\mathbf{A}) = 2.04 \times 10^{18}$ , was an

order of magnitude greater than the maximum value of the effective condition number,  $\eta(\mathbf{A}, \mathbf{b}) = 1.11 \times 10^{17}$ . This was expected due to the condition number being an overly pessimistic predictor of the sensitivity to noise. The effective condition number increases as increasingly small singular values were added to  $\mathbf{b}$  as expected. However,  $\eta(\mathbf{A}, \mathbf{b})$  plateaus at  $i = 21$  which is explained by the Picard plot shown in Figure 4.3.

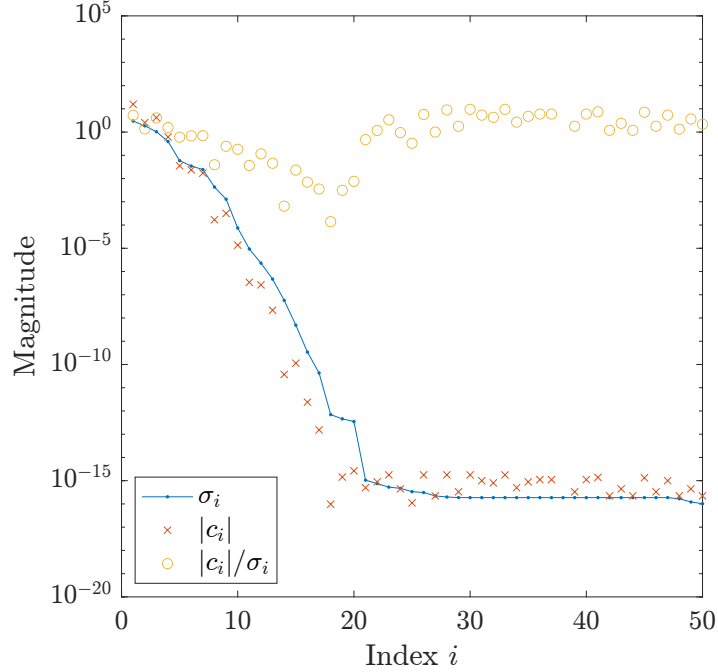


Figure 4.3: Picard plot of the **shaw** problem.

The Fourier coefficients  $|c_i|$  of the **shaw** problem also plateau at around  $i = 21$  signalling that the floor limit of the double-precision floating-point format has been reached, so adding more left singular vectors to  $\mathbf{b}$  has no effect on  $\eta(\mathbf{A}, \mathbf{b})$ .

#### 4.1.3 Bias-variance Trade-off Visualisation

The test to demonstrate what is meant by the bias-variance trade-off was conducted on the **shaw**(50) problem with a noise level of  $\sigma_{\text{noise}} = 1 \times 10^{-2}$  yielded the following results.

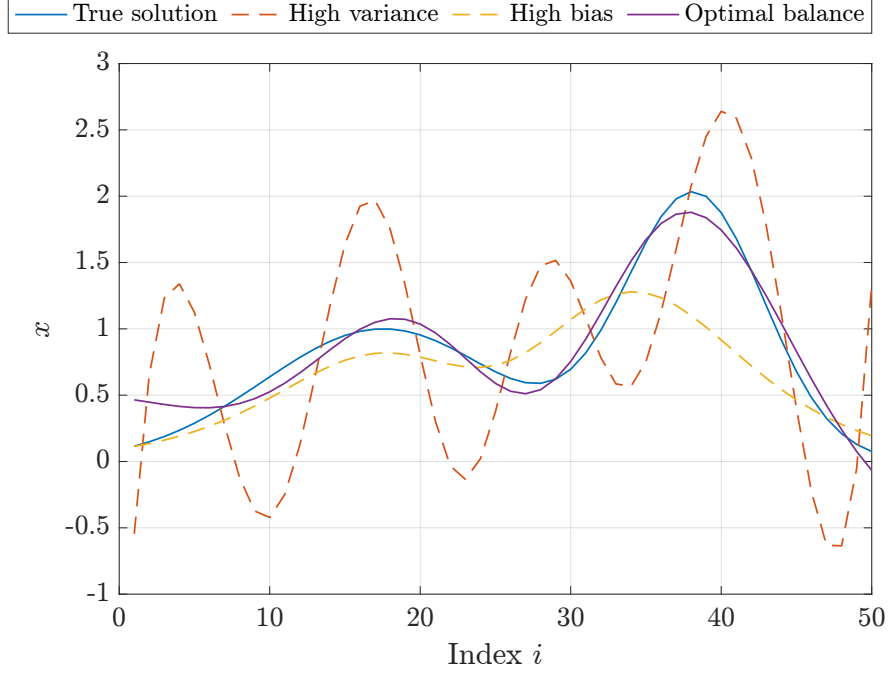


Figure 4.4: Visual representation of the bias-variance trade-off on the **shaw** problem with  $\sigma_{\text{noise}} = 1 \times 10^{-2}$ .

Table 4.2: Numerical comparison of the effect of  $\lambda$  on the relative error, using the **shaw** problem.

| Case            | Relation                   | Regularisation parameter $\lambda$ | $\frac{\ \mathbf{x}_{\text{true}} - \mathbf{x}_{\lambda}\ _2}{\ \mathbf{x}_{\text{true}}\ _2}$ |
|-----------------|----------------------------|------------------------------------|------------------------------------------------------------------------------------------------|
| High variance   | $0.05\lambda_{\text{opt}}$ | $4.64 \times 10^{-4}$              | 0.7096                                                                                         |
| Optimal balance | $\lambda_{\text{opt}}$     | $9.04 \times 10^{-3}$              | 0.1169                                                                                         |
| High bias       | $100\lambda_{\text{opt}}$  | $9.04 \times 10^{-1}$              | 0.3693                                                                                         |

Figure 4.4 shows four lines representing the true solution, along with three regularised solutions with different regularisation parameters  $\lambda$  corresponding to a high variance, high bias and the optimal trade-off value. The specific values, detailed in Table 4.2, were selected to visually demonstrate the extremes of the regularisation spectrum. The high variance case failed to capture the overall shape due to the low bias and oscillated excessively while the high bias case captured the overall shape correctly, however was too smooth. The optimal balance case, corresponding to the  $\lambda_{\text{opt}}$  chosen by the L-curve shown in Figure 4.9a, managed to capture the overall shape and also the correct magnitudes which is why it is the best choice. Note that it does not exactly match the true solution — this is expected since regularisation solves a neighbouring problem due to the inherent introduction of a small and controlled amount of bias. Table 4.2 shows the relative error corresponding to each case for which the use of the optimal regularisation parameter  $\lambda_{\text{opt}}$  minimises the relative error thus demonstrating the importance of the parameter selection method. Note also that the high bias case

has approximately half the error of the high variance case despite being further away from  $\lambda_{\text{opt}}$  which suggests that over-smoothing is safer than under-smoothing.

#### 4.1.4 Impact of Noise on Spectral Filtering and the DPC

A comparison of the two key analytical regularisation techniques, TSVD and Tikhonov regularisation, was conducted on the **shaw**(50) problem using varying levels of noise to produce the following results.

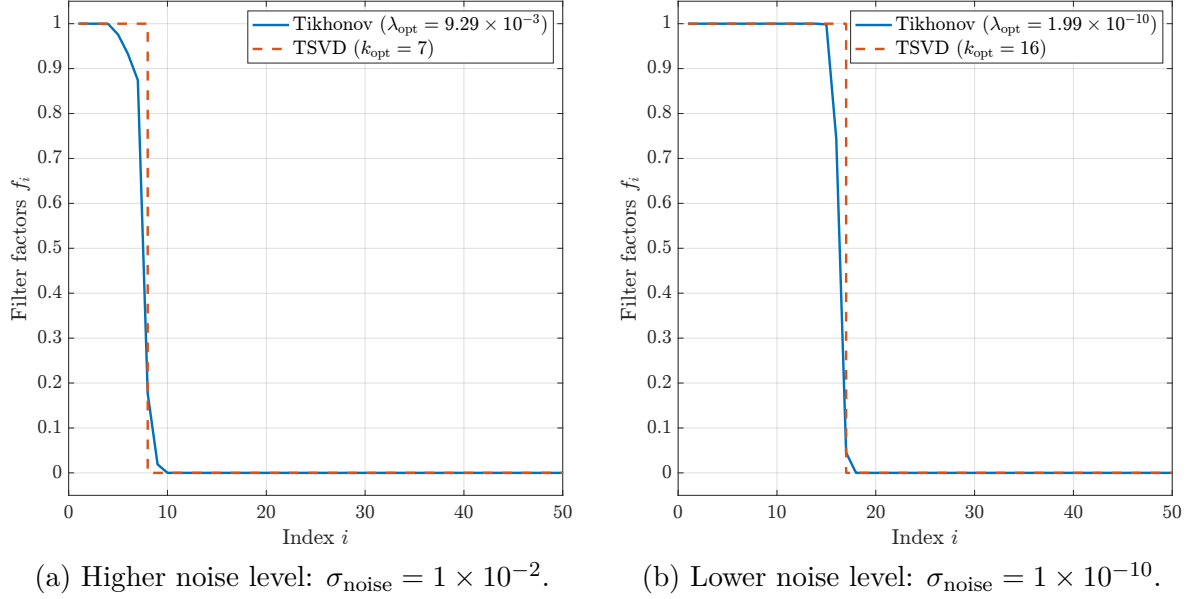


Figure 4.5: Comparison of the filter factors of TSVD and Tikhonov on the **shaw** problem at two different noise levels. (a) Filter factors at higher noise level. (b) Filter factors at lower noise level.

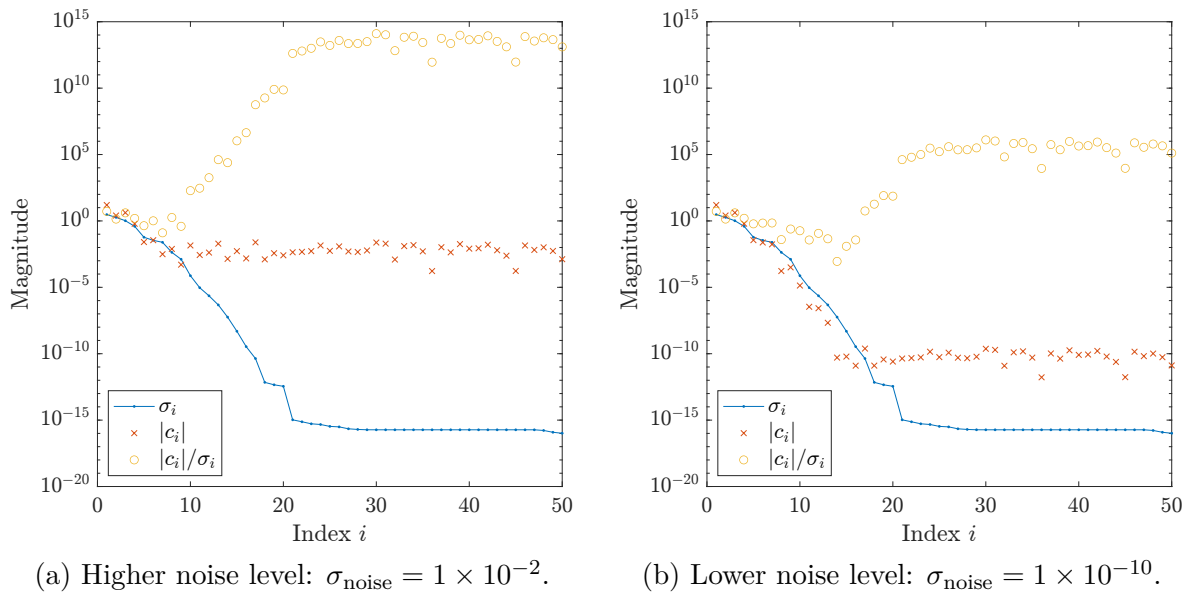


Figure 4.6: Comparison of the Picard plots on the **shaw** problem at two different noise levels. (a) Picard plot at higher noise level. (b) Picard plot at lower noise level.

Figure 4.5 shows the smooth continuous filtering associated with Tikhonov regularisation and the discrete step filtering associated with TSVD as mentioned in Section 2.2. Figure 4.5 also clearly shows that for higher noise levels there are fewer singular values  $\sigma_i$  available to use since the step is further to the left for the higher noise level in Figure 4.5a. However, reducing the noise does not increase  $k_{\text{opt}}$  to up to  $n$  since the limit caused by the double-precision floating-point format is reached at  $k_{\text{opt}} = 21$  for `shaw(50)` with zero noise. It is worth noting that the filter factors for Tikhonov regularisation are not perfectly smooth, however this is solely due to the discretisation level  $n = 50$  so would become increasingly smooth as  $n$  increases. Figure 4.6 shows the associated Picard plots for the filter factors plot in Figure 4.5 and it becomes clear that the index for which the DPC is no longer satisfied lines up exactly with the step in the TSVD — in Figure 4.6a for the first seven indices the Fourier coefficients decay faster than the singular values (DPC satisfied) lining up with the  $k_{\text{opt}} = 7$  shown in Figure 4.5a. These results can be perfectly explained using the following mathematical reasoning.

The DPC requires that the Fourier coefficients  $|c_i|$  decay faster than the singular values  $\sigma_i$ . Let  $|c_i + \delta c_i|$  be the Fourier coefficients in the presence of noise, where  $\delta c_i$  represents the noise contribution. Assume  $\delta c_i \approx \text{constant}$  for all  $i$ . Consider the following ratio,  $\gamma_i$ , the DPC in the presence of noise

$$\gamma_i = \frac{|c_i + \delta c_i|}{\sigma_i}. \quad (4.1)$$

The point at which the magnitude of the Fourier coefficients equals the magnitude of the noise,  $|c_i| \approx |\delta c_i|$ , is the minimum of  $\gamma_i$  — the index at which this intersection occurs is  $i^*$ :

$$\gamma_{i^*} = \frac{|c_i + \delta c_i|}{\sigma_i} \approx \frac{2|c_i|}{\sigma_i}. \quad (4.2)$$

Consider the signal-dominated region ( $i < i^*$ ) where  $|c_i| > |\delta c_i|$ , (4.1) becomes

$$\gamma_i = \frac{|c_i + \delta c_i|}{\sigma_i} \approx \frac{|c_i|}{\sigma_i}. \quad (4.3)$$

Assume the DPC is satisfied in the absence of noise; therefore,  $\gamma_i$  is a decaying function. Now consider the noise-dominated region ( $i > i^*$ ) where  $|\delta c_i| > |c_i|$ , (4.1) becomes

$$\gamma_i = \frac{|c_i + \delta c_i|}{\sigma_i} \approx \frac{|\delta c_i|}{\sigma_i} \quad (4.4)$$

where  $|\delta c_i| \approx \text{constant}$  and  $\sigma_i$  is a decaying function, making  $\gamma_i$  an increasing function. Therefore the presence of noise in theory makes the function  $\gamma_i$  on a Picard plot V-shaped with the minimum at  $i = i^*$  as observed in Figure 4.6. Note that the V-shape levels off around  $i = 21$  because the floor limit of the double-precision floating-point format has been reached, as shown in Figure 4.3. If the noise level increases,  $|\delta c_i|$  becomes larger and so the intersection with the decaying  $|c_i|$  happens at a lower index —  $i^*$  decreases. Note that  $i^*$  is the theoretical limit of information, while  $k_{\text{opt}}$  is the parameter that targets said limit.

#### 4.1.5 Parameter Selection Methods as Diagnostic Tools

This test compared the parameter selection methods, L-curve and GCV, on their ability to diagnose whether regularisation is required or not which is critical for this project. This test was conducted on the mildly ill-posed `deriv2(50)` problem with a fixed noise level,  $\sigma_{\text{noise}} = 1 \times 10^{-2}$  to produce the following results.

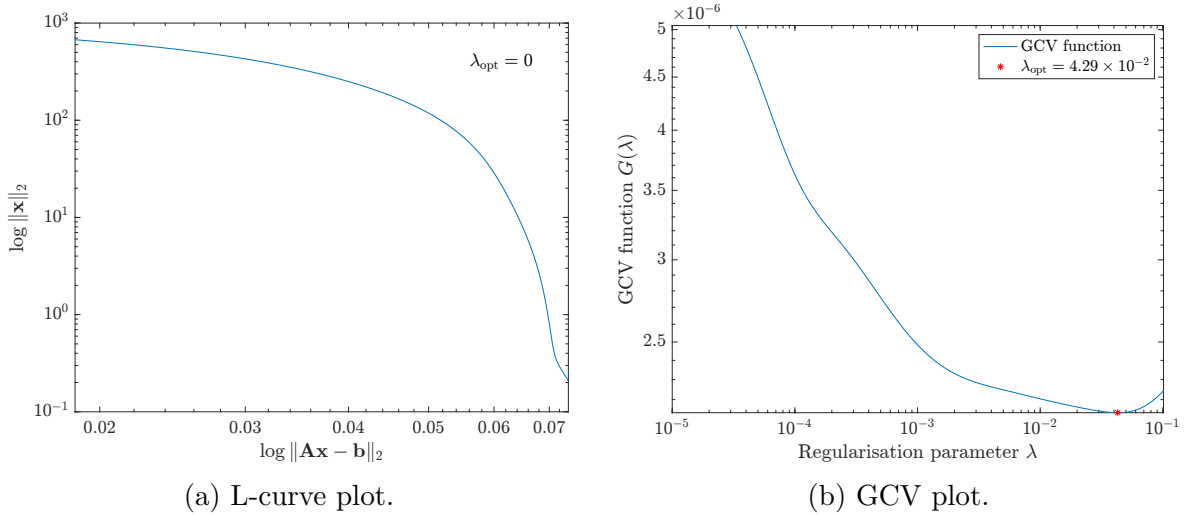


Figure 4.7: Comparison of the L-curve and GCV on the `deriv2` problem. (a) The L-curve shape showing qualitatively not to regularise. (b) The GCV having a minimum, incorrectly indicating the need for regularisation.

The L-curve plot in Figure 4.7a is monotonic and lacks the distinct corner and L-shape that qualitatively indicates that regularisation must not be applied, meaning the regularisation parameter must be set to  $\lambda_{\text{opt}} = 0$ . The L-curve shows that there is no value of  $\lambda$  that minimises both the solution and residual norm simultaneously. It is worth noting that the `l_curve` function does output a  $\lambda_{\text{opt}}$  value, so the `l_curve` function cannot be used quantitatively to determine if regularisation is required blindly. On the other hand, the GCV plot in Figure 4.7b has a relatively deep global minimum at which  $\lambda_{\text{opt}} = 4.29 \times 10^{-2}$  — this incorrectly indicates the need for regularisation both qualitatively and quantitatively. These findings are significant as this determined



the method for assessing whether a real-world machine learning dataset requires regularisation, since often regularisation is applied routinely, introducing unnecessary bias to stable problems — this is due to the inherent nature of regularisation solving a neighbouring problem when the original system is already sufficiently well-posed.

#### 4.1.6 Robustness of Parameter Selection Methods

A Monte Carlo simulation to assess the robustness of the parameter selection methods, L-curve and GCV, was conducted on the `shaw(50)` problem using  $\sigma_{\text{noise}} = 1 \times 10^{-2}$  over 1000 trials. This yielded the following box plot in Figure 4.8 and data shown in Table 4.3.

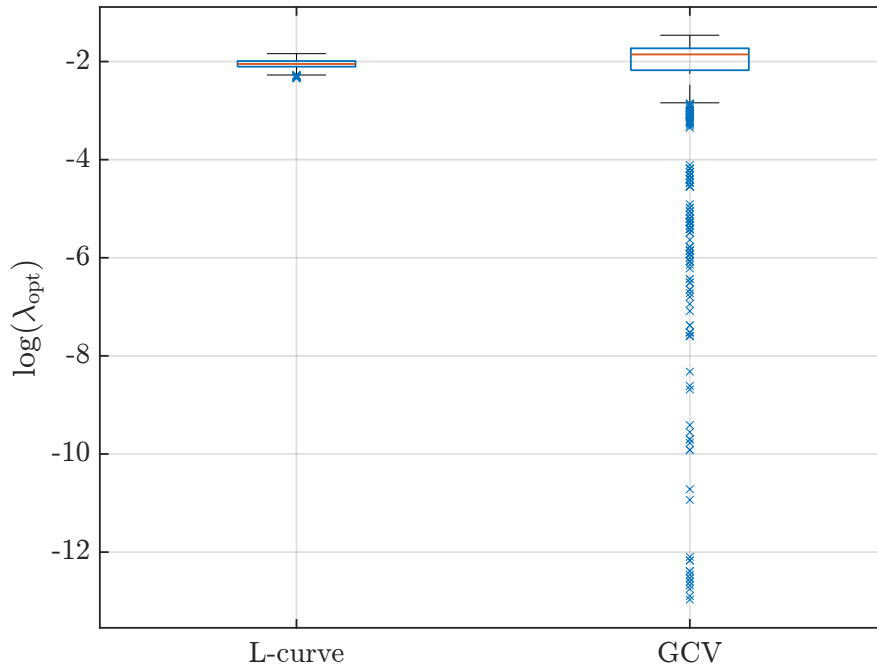


Figure 4.8: Box plot comparison between the L-curve and GCV parameter selection methods on the `shaw` problem over 1000 Monte Carlo trials.

Table 4.3: Statistical robustness analysis of  $\lambda_{\text{opt}}$  parameter choice method over 1000 Monte Carlo trials on the `shaw` problem.

| Method  | Mean $\lambda_{\text{opt}}$ | Standard deviation    | Coefficient of variation | Outliers |
|---------|-----------------------------|-----------------------|--------------------------|----------|
| L-curve | $9.04 \times 10^{-3}$       | $1.74 \times 10^{-3}$ | 19.2%                    | 13       |
| GCV     | $1.31 \times 10^{-2}$       | $8.22 \times 10^{-3}$ | 62.7%                    | 170      |

The box plot for the L-curve, shown in Figure 4.8, has a relatively narrow box and IQR centred around  $\log(\lambda_{\text{opt}}) \approx -2$  with outliers close to the lower whisker. On the other hand, the box plot for the GCV shown in Figure 4.8 has a significantly wider box and IQR with many more outliers, some of which are up to approximately ten orders of magnitude smaller than the lower whisker. This box plot is distilled into

easily understandable data in Table 4.3 which highlights several findings. Firstly, the mean  $\lambda_{\text{opt}}$  value is higher for the GCV compared to the L-curve. Also, the standard deviation and coefficient of variation for the GCV are 4.72 and 3.27 times greater, respectively, than for the L-curve. Finally, there are 170 outliers for the GCV compared to 13 for the L-curve. These results all point towards the fact that the L-curve is a significantly more robust parameter selection method than the GCV. Not only this, but the results also point towards the fact that the GCV is a more conservative parameter selection method than the L-curve because the mean value of  $\lambda_{\text{opt}}$  is higher over the 1000 trials — this means the GCV has a tendency to over-smooth the solution. A possible explanation for the robustness of the L-curve over the GCV can be visualised in Figure 4.9 which was produced using the ill-posed `shaw(50)` problem with a fixed noise level of  $\sigma_{\text{noise}} = 1 \times 10^{-2}$ .

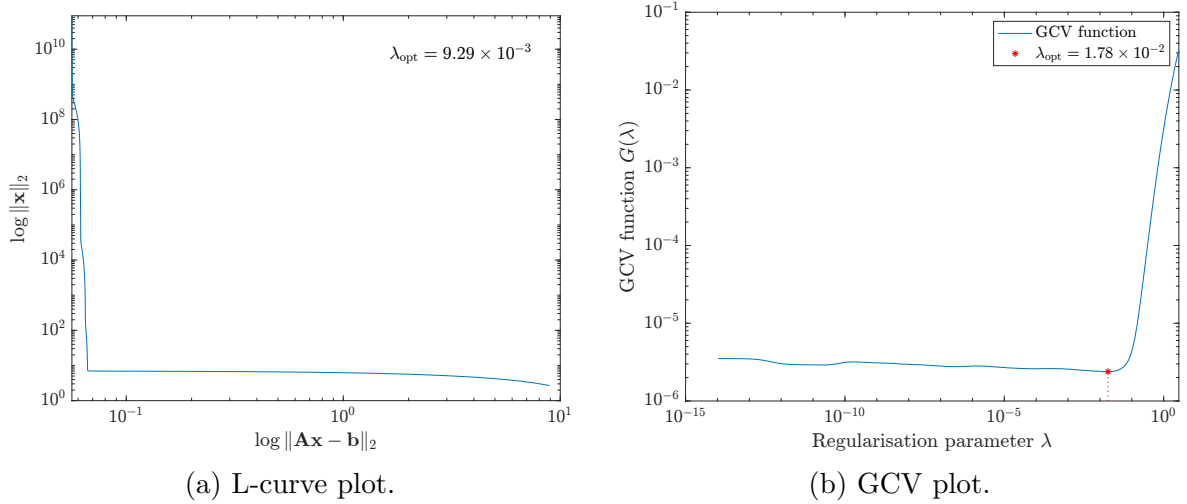


Figure 4.9: Comparison of the L-curve and GCV on the `shaw` problem. (a) The L-curve has a distinct corner, identifying  $\lambda_{\text{opt}}$ . (b) The GCV has a less distinct minimum, identifying  $\lambda_{\text{opt}}$ .

Again the GCV identifies a larger  $\lambda_{\text{opt}}$  than the L-curve, confirming the suspected conservativeness of the GCV. The L-curve plot shown in 4.9a has a sharp, distinct corner identifying  $\lambda_{\text{opt}}$  compared to the shallow minimum identifying  $\lambda_{\text{opt}}$  in the GCV plot shown in 4.9b. This shallow minimum compared to the geometrically distinct L-curve explains why the GCV is much less robust since the  $\lambda_{\text{opt}}$  can change by many orders of magnitude over small changes in  $G(\lambda)$  due to errors — it is highly sensitive.

## 4.2 The Tecator Dataset

### 4.2.1 Tecator Dataset Diagnosis

Before any tests were conducted on the Tecator dataset, it was important to diagnose the problem using the fundamentals of regularisation including the condition number, VIF, DPC, L-curve and GCV.

Table 4.4: Ill-conditioning and multicollinearity diagnostics for the Tecator dataset ( $n = 100$  features).

| Diagnostic metric          | Value                 |
|----------------------------|-----------------------|
| Condition number, $\kappa$ | $1.48 \times 10^7$    |
| Minimum VIF                | $1.85 \times 10^8$    |
| Mean VIF                   | $4.16 \times 10^9$    |
| Maximum VIF                | $1.39 \times 10^{10}$ |

Table 4.4 shows the condition number and an overview of the VIF. The condition number is between that of `shaw`,  $\kappa_{\text{shaw}} = 2.93 \times 10^{16}$ , and `deriv2`,  $\kappa_{\text{deriv2}} = 3.04 \times 10^3$ , indicating it is moderately ill-conditioned. The extremely high minimum, mean and maximum VIF, all being much greater than 10, indicate that there is extreme multicollinearity which is a key characteristic of the Tecator dataset.

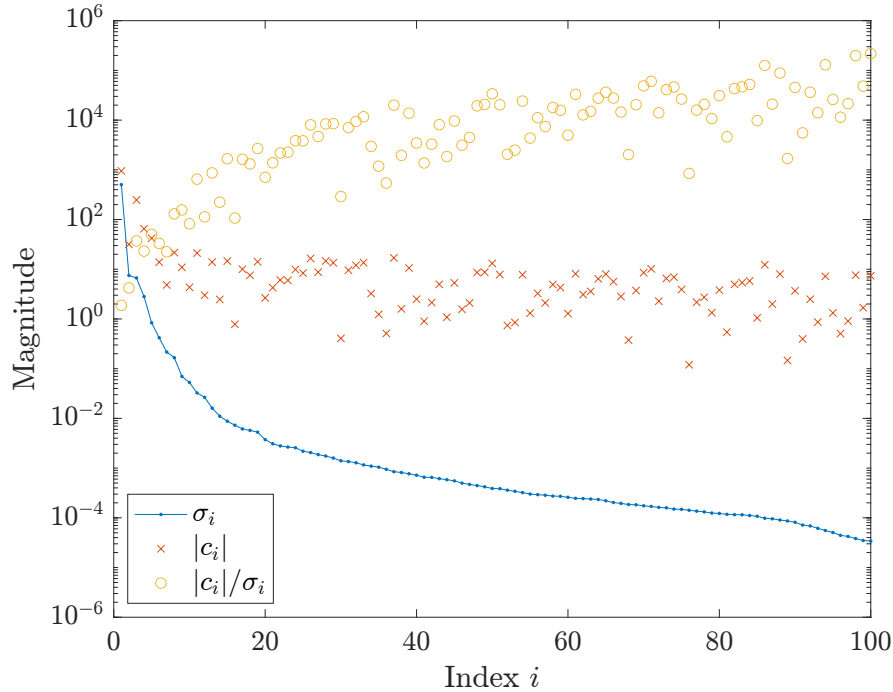


Figure 4.10: Picard plot for the Tecator dataset using moisture as the target.

Figure 4.10 shows the Picard plot for the Tecator dataset for which the DPC is satisfied for all 100 features. This implies that a meaningful regularised solution is possible as

desired. Note the magnitude of the smallest singular value is approximately ten orders of magnitude above that of the ill-posed **shaw** problem, confirming that the Tecator dataset is not as severely ill-posed. The L-curve was also produced, as shown in Figure 4.11.

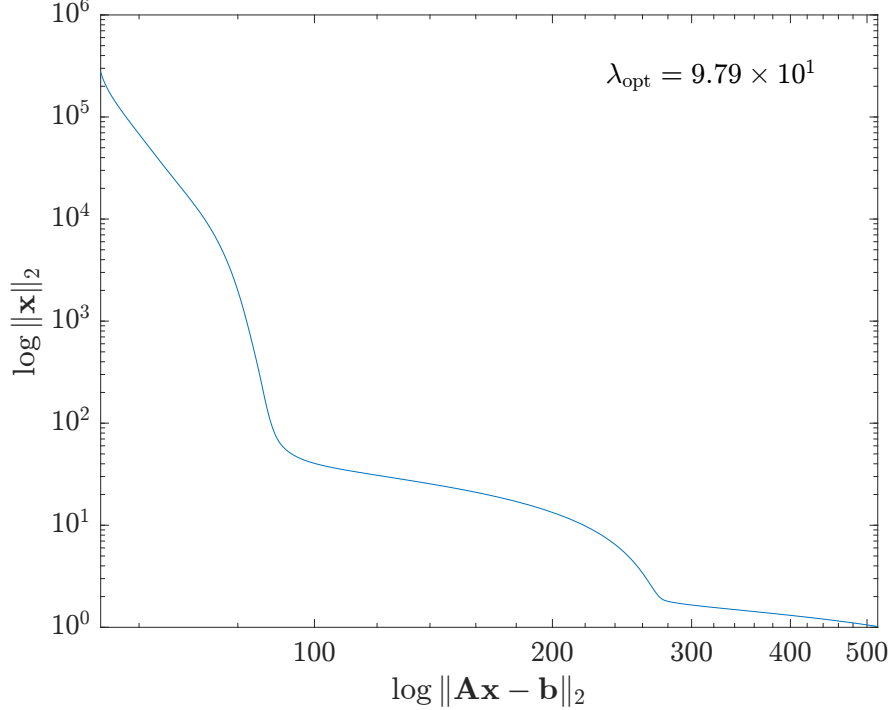


Figure 4.11: L-curve for the Tecator dataset using moisture as the target.

Qualitatively, the L-curve in Figure 4.11 is somewhere in between the L-curve of a severely ill-posed (**shaw**) and mildly-posed problem (**deriv2**) as seen in Figure 4.9a and Figure 4.7a respectively. While not having the most distinct corner, the overall shape indicates that regularisation should be applied — this validated this real-world dataset as being useful for testing. The output of the optimum regularisation parameter using the `l_curve` function of  $\lambda_{\text{opt}} = 9.79 \times 10^1$  was much higher than expected. This is because this point corresponds to the second of two L-shaped kinks which, despite having a higher curvature, is further from the origin. Since both the residual and solution norms should be minimised this indicates that the `l_curve` function has failed to find the correct  $\lambda_{\text{opt}}$ . The GCV plot for the Tecator dataset is shown in Figure 4.12.

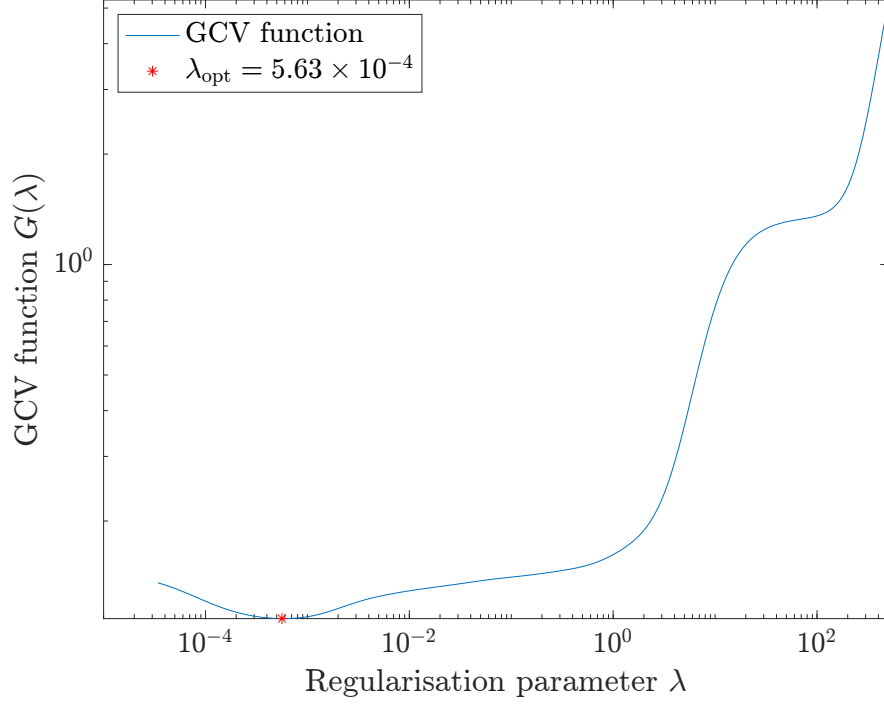


Figure 4.12: GCV plot for the Tecator dataset using moisture as the target.

The GCV plot in Figure 4.12 has a clear minimum which identified the optimum regularisation parameter as  $\lambda_{\text{opt}} = 5.63 \times 10^{-4}$ . The magnitude of  $\lambda_{\text{opt}}$  is as expected unlike for the L-curve. Therefore, the GCV was the chosen method used for identifying the optimum regularisation parameters for the analytical regularisation techniques, Tikhonov regularisation and TSVD. Now the dataset has been diagnosed, testing can begin.

#### 4.2.2 Comparison of Regularisation Techniques on the Tecator Dataset

The key regularisation techniques touched upon were tested on both the full, overdetermined Tecator dataset and a randomly truncated underdetermined version. An 80/20 training/testing data split was used and the results were averaged over 50 trials.

Table 4.5: Performance comparison of regularisation techniques on the full Tecator dataset for moisture prediction with  $\mathbf{A}_{\text{train}}$  as a  $192 \times 100$  matrix, averaged over 50 trials.

| Technique     | Mean RMSE | Std. dev. | Mean features | Run time (s) |
|---------------|-----------|-----------|---------------|--------------|
| Unregularised | 10.1933   | 3.3646    | 100           | 0.12         |
| Tikhonov      | 5.8190    | 0.7430    | 100           | 7.17         |
| TSVD          | 5.9942    | 0.7457    | 100           | 54.61        |
| Elastic net   | 2.9643    | 0.3478    | 60.7          | 1673.67      |
| Lasso         | 2.8949    | 0.3621    | 37.7          | 246.63       |

Table 4.5 shows the mean value of the RMSE, the standard deviation of the RMSE, the mean number of active features and the run time for each of the techniques on the full Tecator dataset. Table 4.5 shows that as expected, the unregularised model has the highest mean RMSE. The analytical techniques, Tikhonov and TSVD, have a very similar mean RMSE, reducing the mean RMSE by approximately 42% from the unregularised model. The statistical techniques, the elastic net and lasso, both reduced the mean RMSE by approximately 71% from the unregularised model, with the lasso slightly outperforming the elastic net whilst using 23 less active features on average and doing so almost seven times quicker. This clearly indicates that the lasso is the best technique for this dataset when overdetermined. The unregularised model runs almost instantly while the analytical techniques run within a minute with Tikhonov being considerably faster than the TSVD. The statistical techniques, the lasso and elastic net, take approximately 4 and 28 minutes, respectively. As expected, the analytical techniques do not shrink any of the features to zero whilst the statistical techniques do. Interestingly, the standard deviation of the RMSE is halved from the analytical to statistical techniques, therefore implying they are more robust to variations in the training data distribution.

Table 4.6: Performance comparison of regularisation techniques on the underdetermined Tecator dataset for moisture prediction with  $\mathbf{A}_{\text{train}}$  as a  $99 \times 100$  matrix, averaged over 50 trials.

| Technique     | Mean RMSE | Std. dev. | Mean features | Run time (s) |
|---------------|-----------|-----------|---------------|--------------|
| Unregularised | 99.6778   | 74.1485   | 95.1          | 0.21         |
| Tikhonov      | 6.1632    | 0.3920    | 100           | 6.09         |
| TSVD          | 6.4127    | 0.7051    | 100           | 19.50        |
| Elastic net   | 3.0836    | 0.2623    | 66            | 2600.51      |
| Lasso         | 3.0765    | 0.2143    | 32.6          | 325.83       |

Table 4.6 is the same as seen in Table 4.5, but for the underdetermined case. The mean RMSE for each technique is slightly higher than for the overdetermined case in Table 4.5 except for the unregularised case which has a mean RMSE an order of magnitude higher, that rounds to 100. This means it is completely unusable as it is numerically unstable. This is because the pseudoinverse for  $m < n$  shown in Section 2.1 is rank-deficient so there are infinite solutions. The mean RMSE is slightly higher for the underdetermined dataset because the training dataset is almost halved so there is less data, and mean RMSE decreases when more training data is available. It is worth noting that though the mean RMSE across the board is higher, the standard deviation of the RMSE is lower except for the outlier which is the unregularised model. This is because the underdetermined case has 93 more unseen testing points than the full dataset, reducing the impact of individual outliers on the mean RMSE. While the

gap between the mean RMSE between the lasso and elastic net has reduced compared to the overdetermined case, the lasso still performs best, using less features and significantly quicker than for the elastic net. This was not expected since the findings by Zou & Hastie (2005) indicated that the elastic net outperforms the lasso when  $m < n$  or when there is high multicollinearity — this shows how incredibly redundant NIR data is. The run time for the analytical techniques is similar for over and underdetermined, however for the statistical techniques, particularly the elastic net, the run time increased significantly (15 minutes longer run time for elastic net) for the underdetermined case.

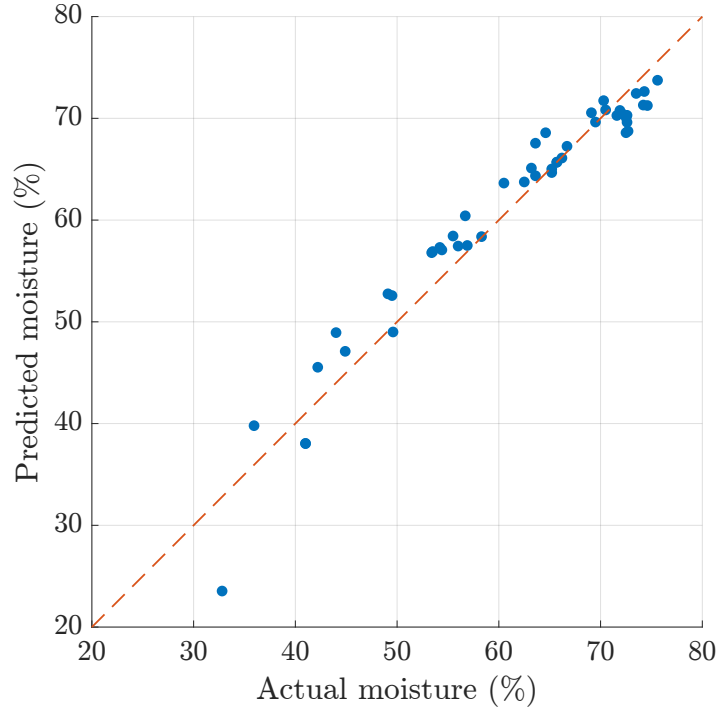


Figure 4.13: Lasso regression plot for the Tecator dataset for moisture prediction with  $\mathbf{A}_{\text{train}}$  as a  $192 \times 100$  matrix, for the median trial.

Figure 4.13 shows the median trial of the technique with the lowest mean RMSE of 2.8949 which is indicated by the test points fitting closely with the ideal fit, the line  $y = x$ . The error is highest at low levels of moisture, below 35% where lasso underpredicts the value. Also at test points above an actual moisture level of approximately 70%, lasso again underpredicts the value but to a lesser extent. In the middle, lasso tends to slightly overpredict the moisture content. This conservatism at the extremes is characteristic of the shrinkage effect in lasso, where coefficients are constrained to ensure overall model stability. Overall, the lasso successfully arrived at a good bias-variance trade-off with the test points deviating on average 2.8949% from the actual moisture levels.

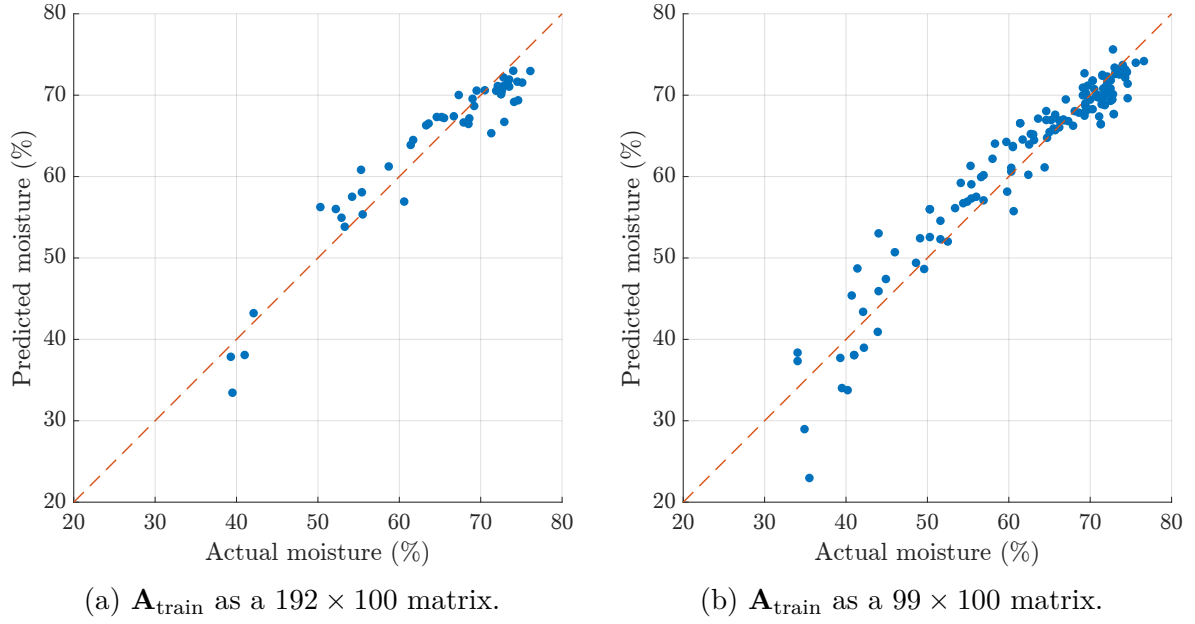


Figure 4.14: Regression plot comparison of the median trial of the elastic net on the Tecator dataset. (a) Full dataset. (b) Underdetermined case.

Figure 4.14 illustrates that the mean RMSE for the underdetermined case is higher since the test points are not as tight around the ideal fit line. Note that the number of test points for the underdetermined and full dataset are 141 and 48 respectively. Similarly for the lasso in Figure 4.13, the test points perform worst at low moisture levels with one test point underpredicting moisture content by approximately 12%. Both plots show that elastic net, like the lasso, generally tends to underpredict at the extremes and overpredict in the middle.



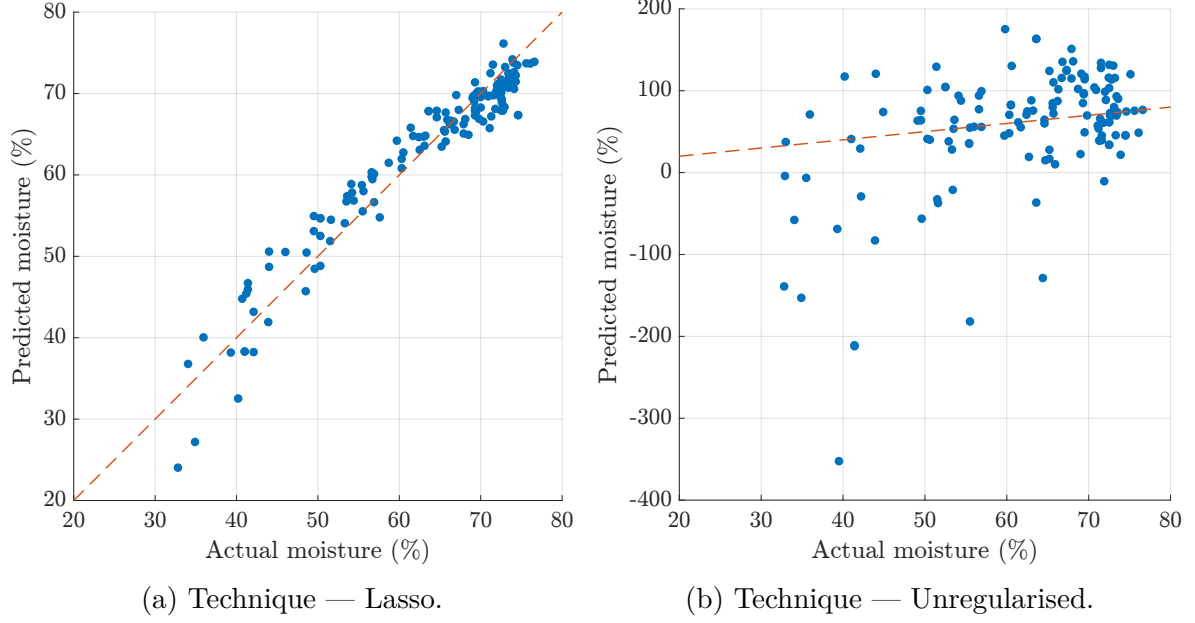


Figure 4.15: Comparison of the median trial of the unregularised and lasso on the underdetermined Tecator dataset. (a) Lasso regression plot. (b) Unregularised regression plot.

Figure 4.15 shows the lasso has a strong linear correlation close to the ideal fit line despite the underdetermined constraint, while the unregularised model has completely broken down which is visual evidence of the numerical instability caused by the rank-deficient  $\mathbf{A}_{\text{train}}$ . The unregularised model predicts moisture contents that are physically impossible since the moisture content must be between 0 and 100%, showing the necessity to regularise this dataset due to the ill-conditioned nature and multicollinearity. Note that the axes labels are no longer matched as they were in Figure 4.13 and Figure 4.14.

#### 4.2.3 Feature Selection with the Lasso on the Tecator Dataset

The following test was conducted to determine how often each of the 100 features, 850 nm to 1050 nm in 2 nm increments, was selected by the lasso using the full Tecator dataset with an 80/20 training/testing split over 50 trials shuffling the training data.

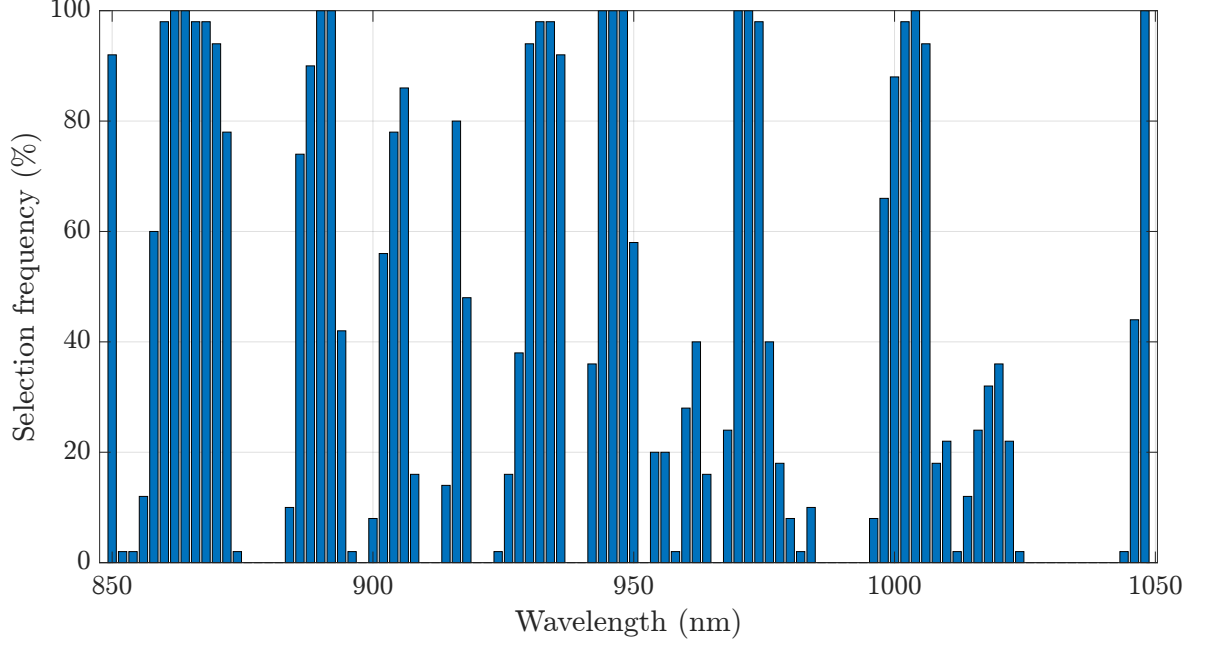


Figure 4.16: Feature selection frequency for the lasso on the full Tecator dataset for moisture prediction with  $\mathbf{A}_{\text{train}}$  as a  $192 \times 100$  matrix.

Ideally there would only be peaks with 100% selection frequency which would indicate robustness but this is not realistic due to the extreme multicollinearity, a mean VIF of  $4.16 \times 10^9$ , identified in Table 4.4. In reality, Figure 4.16 shows that the lasso picked 11 features 100% of the time with other supporting features at lower selection frequencies. Therefore, on average 26.7 features of the 37.7 total features are supporting features that are not always selected. A primary wavelength selection at 970 nm was selected 100% of the time which corresponds exactly to the known NIR range liquid water absorption band at 970 nm (Curcio & Petty 1951) due to the second overtone of the O-H stretching band — this serves as an empirical validation for the model since the target is moisture content. The supporting features, with selection frequency between 0 and 100%, are present due to the high multicollinearity of the Tecator dataset.

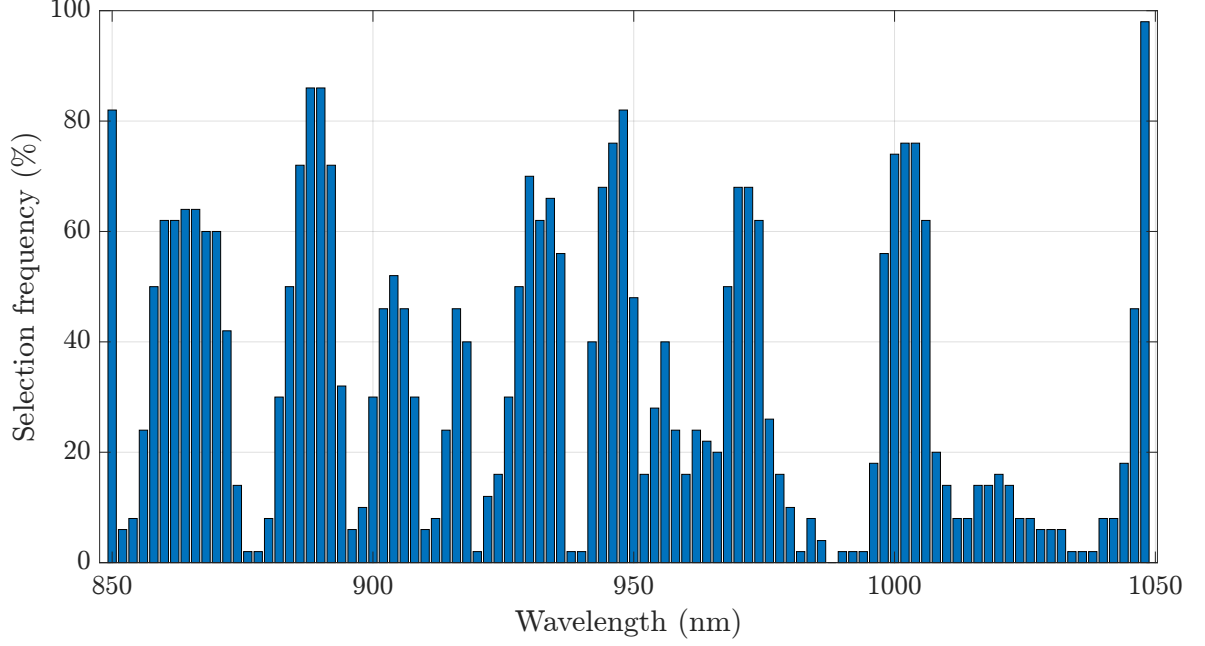


Figure 4.17: Feature selection frequency for the lasso on the underdetermined Tecator dataset for moisture prediction with  $\mathbf{A}_{\text{train}}$  as a  $99 \times 100$  matrix.

While Figure 4.16 identified 11 features that were selected 100% of the time, Figure 4.17 has zero features that are always selected. The 970 nm peak has a reduced selection frequency of approximately 68% which indicates that the physical markers of moisture are more difficult to identify when the system is rank-deficient. Overall, this signifies a reduction in the model stability when moving from overdetermined to underdetermined. The lack of grouping effect with the lasso when underdetermined or there is high multicollinearity, both in this case, is clear from the flickering of selection frequencies across adjacent wavelengths.

#### 4.2.4 Model Convergence with the Lasso on the Tecator Dataset

The lasso was trained with  $m_{\text{train}} \in \{50, 75, 99, 125, 150, 175, 192\}$  training rows and the mean RMSE on the remaining testing rows over 50 trials was recorded to plot what is known as a learning curve.

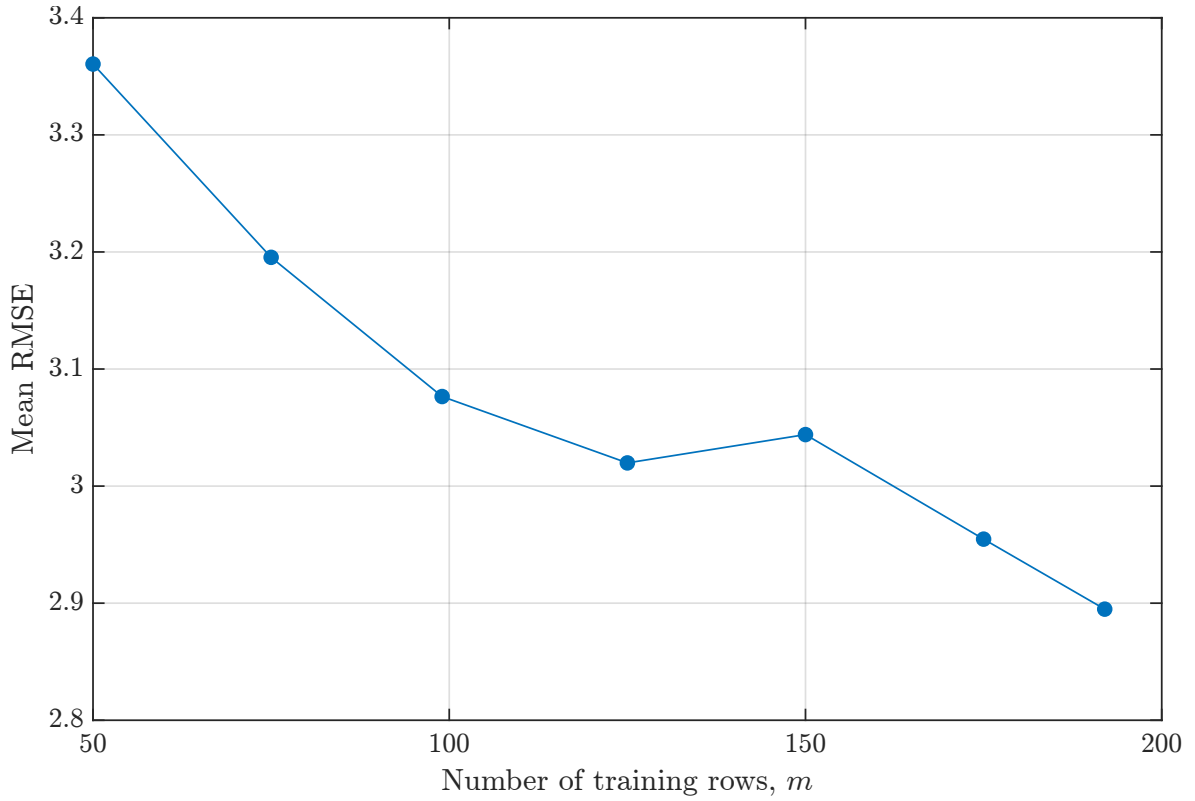


Figure 4.18: Learning curve for the lasso on the Tecator dataset for moisture prediction.

The general trend in Figure 4.18 is that the mean RMSE on unseen testing data decreases as the number of training rows increases. Note that as the line crosses over 100 training rows, the training matrix goes from underdetermined to overdetermined. The fact that the order of magnitude of the mean RMSE is very consistent and the mean RMSE changes with number of training rows builds trust in the dataset. The fact that the mean RMSE does not converge indicates that the dataset would benefit from having more data available which is limiting the mean RMSE, not the regularisation technique. Therefore, further improvements in mean RMSE could be achieved with an expanded dataset.

## 4.3 The California Housing Dataset

### 4.3.1 California Housing Dataset Diagnosis

The first step was as always to diagnose the dataset. To do this, the condition number, VIF and L-curve were used to obtain following results in Table 4.7 and Figure 4.19.

Table 4.7: Ill-conditioning and multicollinearity diagnostics for the full California housing dataset ( $m = 18555$  features).

| Diagnostic metric          | Value              |
|----------------------------|--------------------|
| Condition number, $\kappa$ | $1.95 \times 10^2$ |
| Minimum VIF                | 1.01               |
| Mean VIF                   | 5                  |
| Maximum VIF                | 9.45               |

Table 4.7 shows the condition number, and an overview of the VIF. The condition number is an order of magnitude smaller than that of `deriv2`,  $\kappa_{\text{deriv2}} = 3.04 \times 10^3$  which indicates that the dataset is representative of a well-posed problem. The minimum, mean and maximum VIF are all less than 10, indicating that there is not significant multicollinearity unlike the Tecator dataset. The L-curve for the full California housing dataset containing  $m = 18555$  features is shown in Figure 4.19.

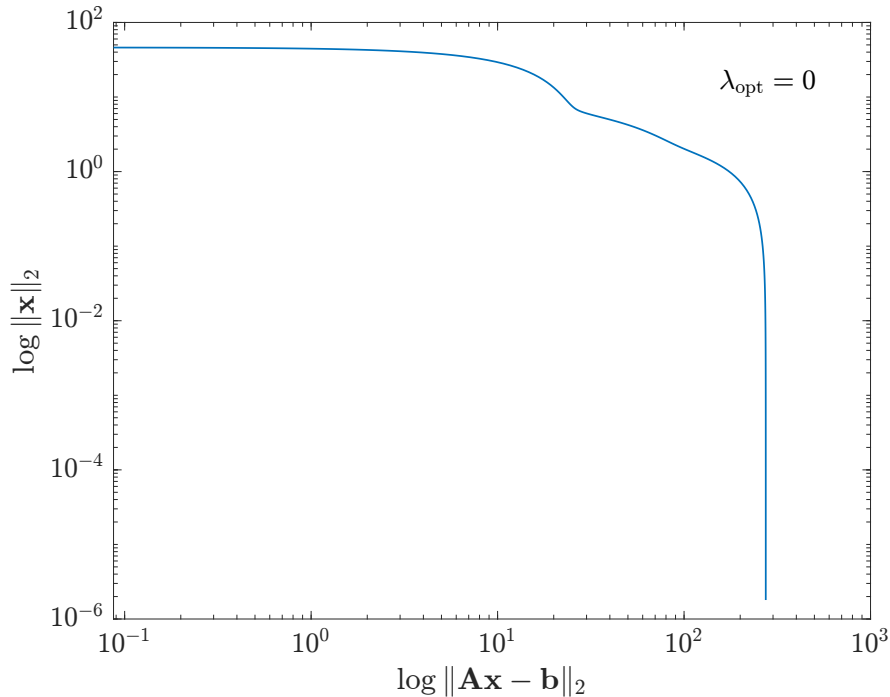


Figure 4.19: L-curve plot for the full California housing dataset ( $m = 18555$  features).

The L-curve in Figure 4.19 has the same shape as seen in Figure 4.7a — it is monotonic and lacks a distinct corner. Qualitatively, this means that regularisation must not be

applied ( $\lambda_{\text{opt}} = 0$ ) because there is no value of  $\lambda$  that simultaneously minimises the solution and residual norm.

#### 4.3.2 Impact of Sample Size on California Housing Dataset

The L-curve of the California housing dataset was plotted using four distinct sample sizes,  $m \in \{100, 1000, 5000, 10000\}$ . To mitigate the impact of sampling bias, a Monte Carlo approach was adopted, where each L-curve represents the mean response over 1000 trials. This process yielded the following results shown in Figure 4.20.

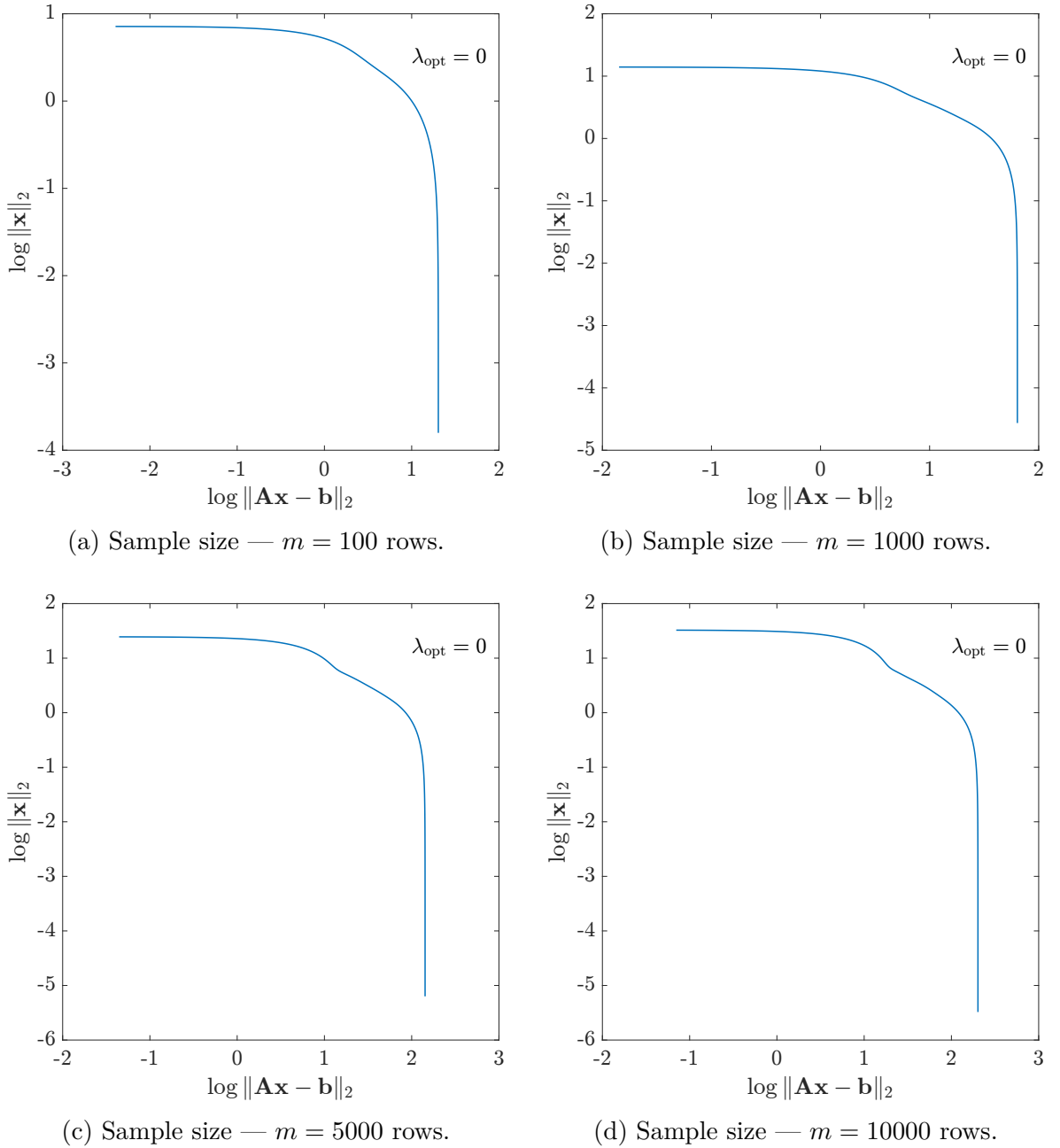


Figure 4.20: L-curve plots for the California Housing dataset across varying sample sizes  $m$ . (a)  $m = 100$ , (b)  $m = 1000$ , (c)  $m = 5000$ , and (d)  $m = 10000$ .

Figure 4.20 shows four graphs that have the same shape: they are monotonic and lack a distinct corner. The significance of the fact they are all the same shape is that this means the dataset is high quality and reliable since it does not matter how many or which samples are chosen — this builds confidence in the results. The shape of the graphs themselves confirm what was found in Figure 4.19, that regularisation must not be applied.

## 5 CONCLUSION

### 5.1 Summary of Findings

This project systematically evaluated the theoretical foundations and practical applications of regularisation in machine learning. By conducting empirical testing on both synthetic problems and real-world datasets described in Section 3, the following key findings summarising Section 4 were established:

- The rule of thumb regarding the condition number  $\kappa$  held, though it is conservative as it overpredicts the relative error by an order of magnitude.
- The effective condition number  $\eta$  was a better predictor of how ill-conditioned matrix  $\mathbf{A}$  is because it considers both  $\mathbf{A}$  and  $\mathbf{b}$ , unlike the condition number  $\kappa$ .
- Demonstrations of the bias-variance trade-off showed that the regularisation parameter selection is critical and to conservatively over-regularise was numerically safer and yields a lower relative error than under-regularisation.
- Increasing the noise level decreases the number of indices that satisfy the DPC. The machine epsilon  $\epsilon_{\text{mach}}$  of the double-precision floating-point format acted as the limit on increasing the number of indices satisfying the DPC when noise is eliminated.
- The L-shaped L-curve is the mathematical condition that signalled regularisation is necessary. Conversely, the lack of a distinct corner signalled that regularisation must not be applied. Here, “must” was used because applying regularisation when it is not required unnecessarily introduces error and can suppress real features. The L-curve is only a qualitative diagnostic tool since the `l_curve` function still outputs a non-zero value of  $\lambda_{\text{opt}}$ , indicating to apply regularisation, since there is always a point of maximum curvature.
- The L-curve can be used as a diagnostic tool to determine whether regularisation is required; however, the GCV could not, as it always qualitatively and quantitatively implied to apply regularisation.
- The L-curve was more robust than the GCV as a parameter selection method. This is because the GCV had a shallow minimum highlighting  $\lambda_{\text{opt}}$  unlike the sharp corner for the L-curve. The GCV consistently outputs a higher value of  $\lambda_{\text{opt}}$  than the L-curve, therefore the GCV is more conservative.
- The L-curve signalled to regularise the highly multicollinear and moderately ill-conditioned Tecator dataset, for which statistical regularisation techniques



(lasso and elastic net) significantly outperformed analytical techniques (TSVD and Tikhonov). The lasso performed better than the elastic net despite using on average 23 fewer features and running seven times quicker on the full dataset. On an underdetermined version of the Tecator dataset, the lasso continued to outperform the elastic net which contradicts the findings from Zou & Hastie (2005). The unregularised case on this underdetermined dataset was entirely dominated by noise.

- The lasso consistently selected 11 out of the 100 features of the full Tecator dataset 100% of the time. The most notable of which was the 970 nm wavelength which corresponds exactly to the known physical NIR absorption band of liquid water, validating the model.
- The learning curve of the lasso on the full Tecator dataset failed to converge which suggests that the mean RMSE of the model would continue to decrease as the training data increased.
- The L-curve signalled that regularisation must not be applied to the well-posed California housing dataset highlighting the danger of treating regularisation as a “black-box” default in machine learning pipelines. The L-curve for different sample sizes told the same story which suggested that the dataset was reliable.

## 5.2 Future Work

The findings of this project highlight the danger of treating regularisation as a “black-box” process, a treatment that is commonly applied by default in machine learning pipelines. Research identified that the lasso is typically blindly applied using Python libraries without understanding why, or stopping to question if the intervention is even required. As established previously, the application of regularisation is not benign, as it unnecessarily introduces error to a solution for an already stable system, which can suppress real features.

A proposed solution to this common problem in machine learning is an automated diagnostic step. In this framework, the user would input  $\mathbf{A}$  and  $\mathbf{b}$  and the algorithm would output a boolean value indicating whether regularisation is necessary. This would remove all ambiguity since the answer is often unclear, as seen in Figure 4.11. Currently the `l_curve` function developed by Hansen (2007) operates by identifying the value of  $\lambda_{\text{opt}}$  at the point of maximum curvature. Since every L-curve has a point of maximum curvature, this always returns a non-zero value of  $\lambda_{\text{opt}}$  which is misleading. This proposed function would calculate  $\kappa(\mathbf{A})$ ,  $\eta(\mathbf{A}, \mathbf{b})$  and the VIF values. Crucially, it would feature a localised corner-finding algorithm that evaluates both the maximum

curvature and the minimal Euclidean distance to the origin. Using primarily the Euclidean distance to the origin and the effective condition number  $\eta(\mathbf{A}, \mathbf{b})$ , this boolean function would determine whether regularisation is required. This would transform the current method, an ambiguous qualitative diagnosis that requires a high baseline level of understanding that most lack, into a clear diagnostic tool.

If regularisation was determined to be necessary, this could extend even further to automating the selection of both the regularisation technique and parameter selection method. Again, this could use  $\mathbf{A}$  and  $\mathbf{b}$  to find the matrices  $\mathbf{U}$ ,  $\mathbf{\Sigma}$  and  $\mathbf{V}$  using the `csvd` function. Also, the system dimensions and the degree of multicollinearity from the VIF values could be determined. The profile of the singular values, the dimensions and degree of multicollinearity would inform the optimal regularisation technique. For example, a gradual singular value decay would prescribe Tikhonov regularisation, a sharp numerical cutoff would prescribe TSVD, many redundant features would prescribe the lasso, and severe multicollinearity or an underdetermined system would prescribe the elastic net.

For the parameter selection method automation, if an analytical technique (Tikhonov regularisation or TSVD) was deemed optimal, the software would use the L-curve or GCV and ensure that the DPC was satisfied up to  $\lambda_{\text{opt}}$ . If a statistical technique (elastic net or lasso) was deemed optimal, the software would implement Bayesian Optimisation as opposed to the K-fold cross-validation approach. This is because K-fold cross-validation typically relies on an exhaustive grid search which has a very high computational cost. On the other hand, Bayesian Optimisation constructs a probabilistic model of the error function where it learns from past evaluations to predict where the lowest error will occur, in a fraction of the time.

All of this could be integrated together into a single function, `regularise`, as demonstrated below: User defines  $\mathbf{A}$  and  $\mathbf{b}$  then calls the function `regularise(A,b)`.

- **If regularisation is required:** the function would build the optimal model by selection of the regularisation technique and parameter selection method and output  $\mathbf{x}_{\text{reg}}$  alongside the RMSE and number of active features.
- **If regularisation is not required:** the function is terminated and the function flags the system as stable, returning the least squares solution  $\mathbf{x}_{\text{ls}}$ .

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## 7 PROJECT MANAGEMENT

An initial project schedule was established in the first semester, detailed in Figure 7.1. Figure 7.2 illustrates the actual project timeline, with the corresponding legend provided in Figure 7.3.

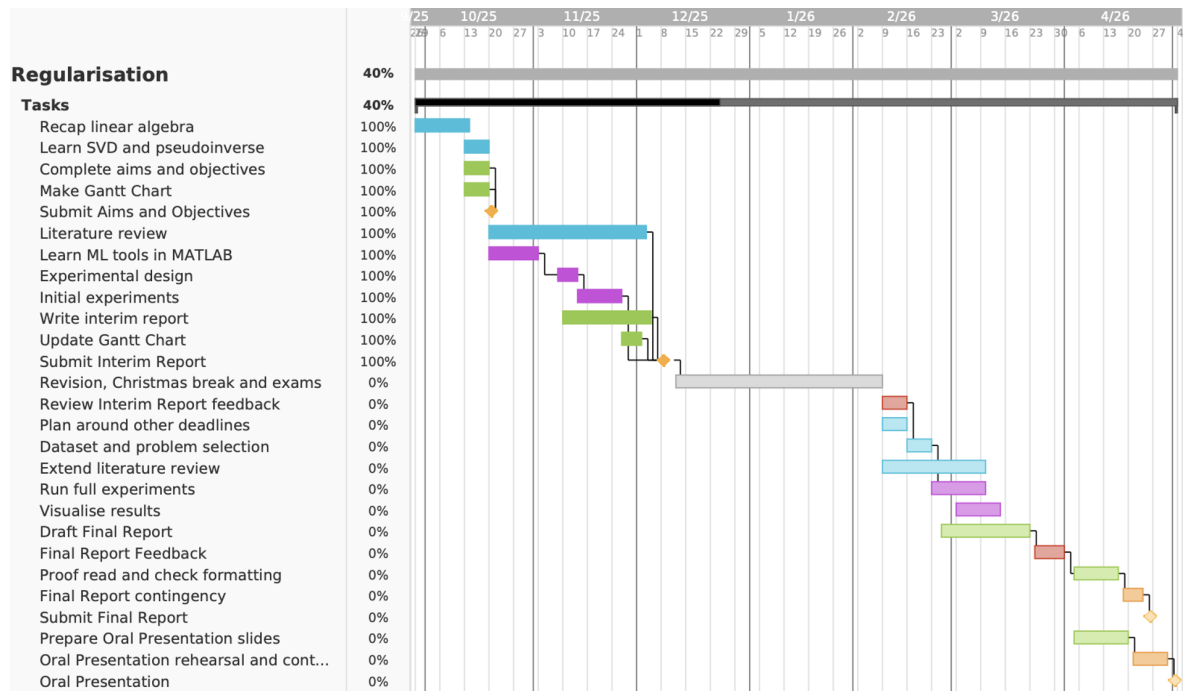


Figure 7.1: Gantt Chart outlined in Interim Report.

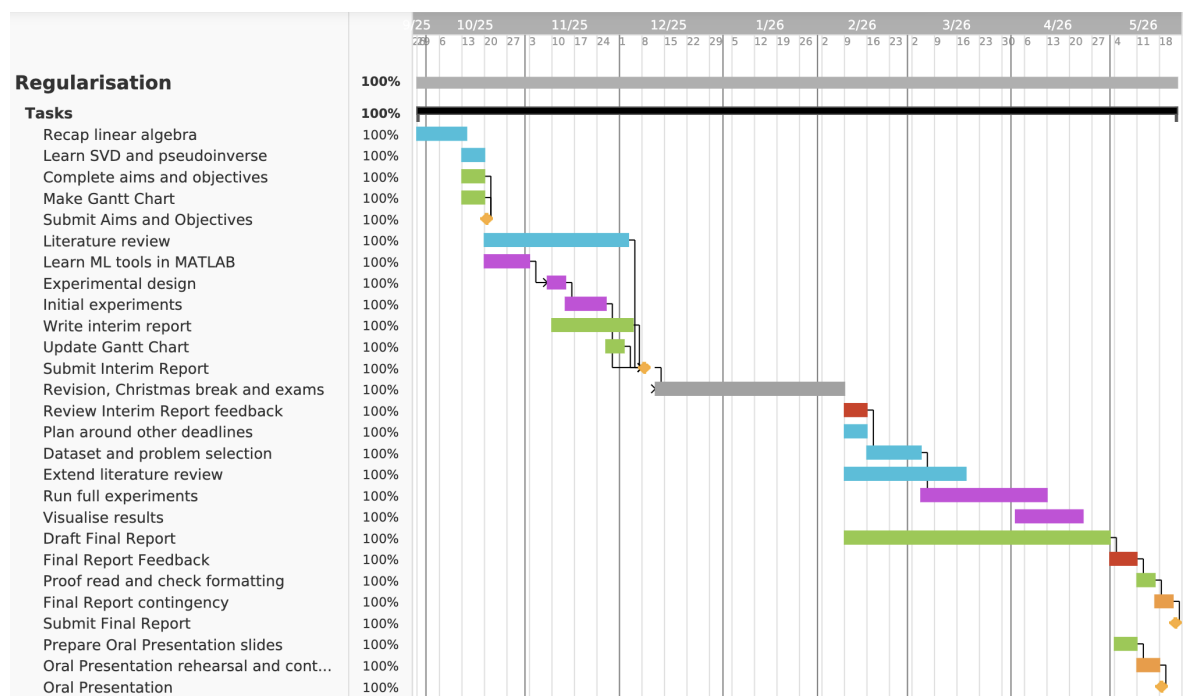


Figure 7.2: Actual project timeline.

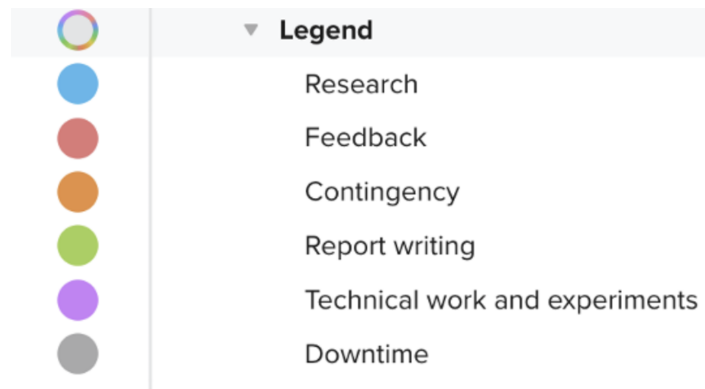


Figure 7.3: Gantt Chart legend.

The Gantt Chart outlined in the Interim Report differed significantly from the actual project timeline. Firstly, the dataset and problem selection took significantly longer due to difficulties finding a real-world ill-posed machine learning dataset. The time taken to extend the literature review was also underestimated in the project plan. Also, running the experiments and visualising the results took longer than expected due to running into many difficulties. These difficulties stemmed from the fact that real-world data rarely behaves as expected and desired. Adapting the project plan and starting the draft as early as possible significantly streamlined the work. As feared, there was less time for proof reading and contingency for the Final Report despite the week extension. The same goes for the oral presentation for which there was less time to prepare than desired. Overall, the project was managed well despite underestimating how long many tasks would take. This is something that will be considered when planning projects in the future to avoid problems.

## 8 SELF-REVIEW

I started this project as a general engineering student, who had specialised in mechanical engineering, with the desire of learning about a new subject area. I had very little knowledge of any of these concepts besides a basic background in linear algebra and I had very limited exposure to statistical machine learning prior to this project. This resulted in a very steep initial learning curve given a lot of these concepts are very challenging. I also had to develop my Matlab skills — this was very successful as I transitioned from initial hesitation to it becoming my preferred programming language. The first semester was largely building my understanding of the background theory required for this project. Once this baseline understanding was established, the project evolved to the application of this theory to real-world datasets which I found very rewarding since I understood the mechanisms behind regularisation.

Overall, I believe that the project aims outlined in Section 1.2 were met successfully. Beyond the core aims, the project significantly broadened my skills including conducting independent research, designing experiments and the considerations involved, and considering how best to present these results.

With more time available, I feel that the scope of this project could have been extended with great success. The finite computational resources available also constrained the project slightly, since the long run times for some tests meant that fewer iterations were used, reducing the sample size and therefore the robustness. Also for testing with the elastic net, I used a discrete set of  $\alpha$  values to reduce the run time so this must be acknowledged. Nevertheless, navigating these constraints was a valuable exercise in resource management and designing the experiments with computation in mind.